

ANGULAR MOMENTUM

The traditional quantum mechanical treatment of central force problems starts from solutions to the time-independent Schrödinger equation, which, for a single particle of mass m moving subject to a potential $V(r)$, is an eigenvalue problem of the general form

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\mathbf{r}) + V(r)\psi(\mathbf{r}) = E\psi(\mathbf{r}). \quad (16.1)$$

Here $\hbar$ is Planck's constant divided by 2π , in SI units approximately 1.05×10^{-34} J-s (joule-seconds); the very small value of this constant causes quantum behavior to be perceptible under most circumstances only at small distances and for particles of small mass; the relevant ranges are typically at atomic scales of mass and length.

The basic interpretation of the Schrödinger equation is that if the energy E of the particle is measured, the result will be one of the eigenvalues of Eq. (16.1), and (subsequent to the measurement) the location of the particle will be described by a probability distribution

$$P(\mathbf{r})d^3r = |\psi(\mathbf{r})|^2d^3r,$$

where $\psi(\mathbf{r})$ is an eigenfunction corresponding to E . As we have seen in Chapters 9 and 15, ψ will in general have angular as well as radial dependence, and its angular part can be written in terms of the spherical harmonics $Y_l^m(\theta, \varphi)$.

A more detailed interpretation of Eq. (16.1) is to identify it as an operator equation in which the momentum $\mathbf{p}$ is identified with the operator $-i\hbar\nabla$, while functions of position, such as the potential energy $V(r)$, are identified as multiplicative operators. Viewed in this way, the operator $-(\hbar^2/2m)\nabla^2$ is seen to represent $p^2/2m$ (i.e., the kinetic energy T), and Eq. (16.1) then becomes equivalent to

$$H\psi \equiv (T + V)\psi = E\psi, \quad (16.2)$$

where H , the Hamiltonian, is an operator whose eigenvalues are the possible values of the total energy.

The Hamiltonian H is a special operator in quantum mechanics because its eigenfunctions yield stationary probability distributions (they do not evolve into different distributions over time). However, H is just like any other quantum operator K representing a

dynamical quantity (with eigenvalues k that can be the result of measurement of K). If ψ is simultaneously an eigenfunction of H and of K , then we can have definite values of both E and k that will not evolve as a function of time, and the measuring of either will not disturb the definite value of the other. This state of affairs can only be achieved if H and K commute, because (see Section 6.4) $[H, K] = 0$ is a necessary and sufficient condition that H and K have a set of simultaneous eigenfunctions.

In earlier chapters, we examined commutators such as $[x, p_x] = i$ (here and except when noted, we use a unit system with $\hbar$ set to unity to avoid unnecessary notational complexity). The nonzero commutator of x and p_x tells us that we cannot simultaneously obtain unambiguous measurement of both these quantities (i.e., we do not have a complete set of states that are simultaneously eigenfunctions of x and p_x). This is the mathematical basis of the Heisenberg uncertainty principle in quantum mechanics.

The notion of simultaneous eigenfunctions and therefore commutation plays a key role in the study of angular momentum in quantum mechanics. Angular momentum is conserved in the classical central force problem, and one of the focal points of the present chapter is to understand the properties of angular momentum operators in quantum mechanics.

16.1 ANGULAR MOMENTUM OPERATORS

In classical physics, the kinetic energy of a particle of mass μ can be written in terms of its momentum $\mathbf{p}$ as $T_{\text{class}} = \mathbf{p}^2/2\mu$. Note that we are using μ for the particle mass to avoid confusion with the usual notation of the azimuthal wave functions φ_m . Most of the literature uses m for both quantities. Introducing spherical polar coordinates, T_{class} can be divided into radial and angular parts, with the angular kinetic energy of the form $\mathbf{L}_{\text{class}}^2/2\mu r^2$. Here $\mathbf{L}_{\text{class}}$ is the angular momentum, defined as $\mathbf{L}_{\text{class}} = \mathbf{r} \times \mathbf{p}$. Following the usual Schrödinger representation of quantum mechanics, the classical linear momentum $\mathbf{p}$ is replaced (in a unit system with $\hbar = 1$) by the **operator** $-i\nabla$. The quantum-mechanical kinetic energy operator is $T_{\text{QM}} = -\nabla^2/2\mu$, which in spherical polar coordinates can be written

$$T_{\text{QM}} = -\frac{1}{2\mu} \left[\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} \right] - \frac{1}{2\mu r^2} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right]. \quad (16.3)$$

Like the classical kinetic energy, T_{QM} can also be divided into radial and angular parts, with the angular part identified in terms of the angular momentum:

$$T_{\text{QM}} = T_{\text{radial, QM}} + \frac{1}{2\mu r^2} \mathbf{L}_{\text{QM}}^2, \quad (16.4)$$

$$T_{\text{radial, QM}} = -\frac{1}{2\mu} \left[\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} \right], \quad (16.5)$$

$$\mathbf{L}_{\text{QM}}^2 = -\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) - \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2}. \quad (16.6)$$

Since our focus here is on the quantum-mechanical operators, we drop the notation “QM” from now on.

The notation $\mathbf{L}^2$ in Eq. (16.6) is only really appropriate if it is consistent with the definition of the quantum-mechanical angular momentum operator, which must have the form

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} = -i\mathbf{r} \times \nabla. \quad (16.7)$$

One way to confirm Eq. (16.6) is to start from the expression for $\mathbf{L}$ in spherical polar coordinates, which can be deduced by applying the operator $\mathbf{r} \times \mathbf{p}$ to an arbitrary function ψ :

$$\mathbf{L}\psi = -i\mathbf{r} \times \nabla\psi = -ir\hat{\mathbf{e}}_r \times \left[\hat{\mathbf{e}}_r \frac{\partial\psi}{\partial r} + \hat{\mathbf{e}}_\theta \frac{1}{r} \frac{\partial\psi}{\partial\theta} + \hat{\mathbf{e}}_\varphi \frac{1}{r \sin\theta} \frac{\partial\psi}{\partial\varphi} \right],$$

from which we extract the formula

$$\mathbf{L} = i \left(\hat{\mathbf{e}}_\theta \frac{1}{\sin\theta} \frac{\partial}{\partial\varphi} - \hat{\mathbf{e}}_\varphi \frac{\partial}{\partial\theta} \right). \quad (16.8)$$

We then rewrite $\mathbf{L}$ in Cartesian components L_x, L_y, L_z (but still expressed in polar coordinates) and evaluate

$$\mathbf{L}^2 = \mathbf{L} \cdot \mathbf{L} = L_x^2 + L_y^2 + L_z^2. \quad (16.9)$$

This process is the topic of Exercise 3.10.32, and leads, as expected, to Eq. (16.6).

In Section 15.5 we identified the solutions of the angular part of the Laplace and Schrödinger equations for central force problems as the spherical harmonics, denoted $Y_l^m(\theta, \varphi)$. Now that we have also written the angular part of these equations in terms of $\mathbf{L}^2$, we see that the Y_l^m can be identified as eigenfunctions of $\mathbf{L}^2$, i.e., that they are angular momentum eigenfunctions, satisfying an eigenvalue equation of the form

$$\mathbf{L}^2 Y_l^m(\theta, \varphi) = l(l+1) Y_l^m(\theta, \varphi). \quad (16.10)$$

Summarizing the discussion to this point, and drawing on previously established properties of the spherical harmonics:

The spherical harmonics Y_l^m are eigenfunctions of $\mathbf{L}^2$ with eigenvalue $l(l+1)$. The eigenfunctions for a given l are $(2l+1)$ -fold degenerate and can be indexed by their m values, which range in unit steps from $-l$ to l .

We now strive for a deeper understanding of the role of angular momentum. The solutions to the time-independent Schrödinger equation are the eigenfunctions of its total energy operator, the Hamiltonian H . We have just observed that for central force problems the angular solutions are eigenfunctions of the angular momentum operator $\mathbf{L}^2$. In order for these two statements to be mutually consistent, it is necessary that H and $\mathbf{L}^2$ commute. For the systems under consideration here, this is clearly true, since we have assumed that H is of the form $T + V(r)$, so

$$H = T_{\text{radial}}(r) + \frac{1}{2\mu r^2} \mathbf{L}^2(\theta, \varphi) + V(r).$$

Since the only angle-dependent quantity in H is the operator $\mathbf{L}^2$, and since $\mathbf{L}^2$ obviously commutes with itself and is independent of r , we have

$$[H, \mathbf{L}^2] = 0.$$

The fact that H and $\mathbf{L}^2$ have simultaneous eigenfunctions in central force problems means that the stationary states of such systems can be characterized by definite values of both the energy and the angular momentum **quantum number** l . States of different l were ultimately identified with series of lines in the emission and absorption spectra of the hydrogen atom that had previously been labeled “sharp,” “diffuse,” “principal,” and “fundamental.” This identification caused physicists to use the initial letters of these names as synonyms for l values; hence it has become essential to know that the code letters for $l = 0, 1, 2$, and 3 are respectively s, p, d , and f . For $l > 3$, the code letters run alphabetically: $g, h, \dots$.

Turning now to the components of $\mathbf{L}$, we have (cf. Exercise 3.10.31)

$$[L_j, L_k] = i\epsilon_{jkn}L_n \quad \text{and} \quad [\mathbf{L}^2, L_j] = 0, \quad (16.11)$$

where j, k, n are different members of the set $(1, 2, 3)$ and ϵ_{jkn} is a Levi-Civita symbol. Although the L_j do not commute with each other, all commute with $\mathbf{L}^2$ and hence also with H , so $H, \mathbf{L}^2$, and any one component of $\mathbf{L}$ mutually commute. We conclude that there exists a set of simultaneous eigenfunctions of $H, \mathbf{L}^2$, and any one component of $\mathbf{L}$. For this purpose we usually pick L_z , motivated by the fact that, in spherical polar coordinates, it is, as found in Exercise 3.10.29,

$$L_z = -i \frac{\partial}{\partial \varphi}. \quad (16.12)$$

For reference, we copy here the far more complicated results for L_x and L_y , obtained from Exercise 3.10.30:

$$\begin{aligned} L_x &= i \sin \varphi \frac{\partial}{\partial \theta} + i \cot \theta \cos \varphi \frac{\partial}{\partial \varphi}, \\ L_y &= -i \cos \varphi \frac{\partial}{\partial \theta} + i \cot \theta \sin \varphi \frac{\partial}{\partial \varphi}. \end{aligned} \quad (16.13)$$

The spherical harmonics are, in fact, eigenfunctions of L_z . Since

$$L_z e^{im\varphi} = -i \frac{\partial}{\partial \varphi} e^{im\varphi} = m e^{im\varphi}, \quad (16.14)$$

we see that Y_l^m is an eigenfunction of L_z with eigenvalue m . This is one of the reasons why the complex exponentials, rather than the trigonometric functions, were chosen in the definitions of the spherical harmonics. It is obvious that $\cos m\varphi$ is not an eigenfunction of L_z : $-i(\partial/\partial\varphi)\cos m\varphi = im\sin m\varphi$. Note, however, that $\exp(\pm im\varphi)$, $\cos m\varphi$, and $\sin m\varphi$ are all eigenfunctions of the operator $L_z^2 = -\partial^2/\partial\varphi^2$ with eigenvalue m^2 .

Ladder Operators

The commutators of the angular momentum components permit the development of some useful algebraic relationships. While these relationships can be found from the specific forms of the operators (cf. Exercise 3.10.30), more general and valuable results are

obtained by derivations based only on the commutators given in Eq. (16.11). We define the operators

$$L_+ = L_x + iL_y, \quad L_- = L_x - iL_y, \quad (16.15)$$

and consider the commutators

$$[L_z, L_+] = [L_z, L_x] + i[L_z, L_y] = iL_y + i(-iL_x) = L_+, \quad (16.16)$$

$$[L_z, L_-] = [L_z, L_x] - i[L_z, L_y] = iL_y - i(-iL_x) = -L_-. \quad (16.17)$$

We start by applying Eq. (16.16) to a function ψ_l^m , which is assumed to be a normalized simultaneous eigenfunction of $\mathbf{L}^2$, with eigenvalue λ_l , and of L_z , with eigenvalue m ; the form of ψ_l^m (and even the space within which it resides) need not be specified to carry out the present discussion. Moreover, at this point we introduce no information about the possible values of λ_l and m . However, to visualize what we are doing, the reader can keep in mind that one possible interpretation of ψ_l^m is the spherical harmonic Y_l^m . We have

$$[L_z, L_+] \psi_l^m = L_z L_+ \psi_l^m - L_+ L_z \psi_l^m = L_+ \psi_l^m.$$

Since $L_z \psi_l^m = m \psi_l^m$, we can rewrite the central and right-hand members of the above equation as

$$L_z (L_+ \psi_l^m) - m (L_+ \psi_l^m) = (L_+ \psi_l^m),$$

which rearranges to

$$L_z (L_+ \psi_l^m) = (m + 1) (L_+ \psi_l^m). \quad (16.18)$$

This tells us that if $L_+ \psi_l^m$ is nonzero, it is an eigenfunction of L_z with eigenvalue $m + 1$; for that reason L_+ can be called a **raising operator**. By itself, this analysis tells us nothing about the value(s) of m , but only that L_+ increases m in unit steps. A similar development shows that L_- is a **lowering operator**, corresponding to the equation

$$L_z (L_- \psi_l^m) = (m - 1) (L_- \psi_l^m). \quad (16.19)$$

Raising and lowering operators are collectively referred to as **ladder operators**.

Next, we recall that $[\mathbf{L}^2, L_i] = 0$ for all components L_i . This means also that $[\mathbf{L}^2, L_+] = 0$, so

$$\mathbf{L}^2 (L_+ \psi_l^m) = L_+ \mathbf{L}^2 \psi_l^m = \lambda_l (L_+ \psi_l^m),$$

showing that $(L_+ \psi_l^m)$ is still an eigenfunction of $\mathbf{L}^2$ with the same eigenvalue, λ_l , as ψ_l^m . Note that we did not need to know the value of λ_l to draw this conclusion. Summarizing, the operators $L_\pm$ convert ψ_l^m into quantities proportional to $\psi_l^{m\pm 1}$, with the conversion failing only if $L_\pm \psi_l^m = 0$.

While Eqs. (16.18) and (16.19) tell us that $L_\pm$ are ladder operators, they do not tell us whether the quantities $L_\pm \psi_l^m$ are normalized. To address this problem, we write the normalization expression for $L_+ \psi_l^m$ in the form

$$\langle L_+ \psi_l^m | L_+ \psi_l^m \rangle = \langle \psi_l^m | L_- L_+ | \psi_l^m \rangle,$$

where we have used the fact that, because L_x and L_y are Hermitian, $(L_+)^{\dagger} = L_-$.

To obtain more information about L_-L_+ , we rearrange $\mathbf{L}^2$ as follows:

$$\mathbf{L}^2 = \frac{1}{2}(L_+L_- + L_-L_+) + L_z^2 = L_-L_+ + \frac{1}{2}[L_+, L_-] + L_z^2, \quad (16.20)$$

a result that can be easily verified by expanding L_+ , L_- , and the commutator. Then we introduce

$$[L_+, L_-] = [L_x + iL_y, L_x - iL_y] = -i[L_x, L_y] + i[L_y, L_x] = 2L_z, \quad (16.21)$$

and solve Eq. (16.20) for L_-L_+ , obtaining

$$L_-L_+ = \mathbf{L}^2 - L_z^2 - L_z. \quad (16.22)$$

Using the fact that ψ_l^m is a normalized eigenfunction of both $\mathbf{L}^2$ and L_z , we can now perform the evaluation

$$\begin{aligned} \langle L_+\psi_l^m | L_+\psi_l^m \rangle &= \langle \psi_l^m | L_-L_+ | \psi_l^m \rangle = \langle \psi_l^m | \mathbf{L}^2 - L_z^2 - L_z | \psi_l^m \rangle \\ &= \lambda_l - m^2 - m. \end{aligned} \quad (16.23)$$

A parallel analysis leads to the companion result

$$\langle L_-\psi_l^m | L_-\psi_l^m \rangle = \langle \psi_l^m | L_+L_- | \psi_l^m \rangle = \lambda_l - m^2 + m. \quad (16.24)$$

If we use the expressions in Eqs. (16.23) and (16.24) to account for the scale factors generated by the ladder operators, we can summarize their action as

$$\begin{aligned} L_+\psi_l^m &= \sqrt{\lambda_l - m(m+1)}\psi_l^{m+1}, \\ L_-\psi_l^m &= \sqrt{\lambda_l - m(m-1)}\psi_l^{m-1}, \end{aligned} \quad (16.25)$$

where, the reader may recall, λ_l is the eigenvalue of $\mathbf{L}^2$ corresponding to quantum number l ; the current analysis has not yet determined its value. The expressions in Eq. (16.25) have also incorporated the assumption that the signs of the ψ_l^m are related as shown. That is a matter of definition, and when the ψ_l^m are taken to be the spherical harmonics Y_l^m , the Condon-Shortley phase assignment was deliberately designed to make Eq. (16.25) consistent with the signs given the Y_l^m in Table 15.4.

Next, we return to Eq. (16.23) and note that since it describes a normalization integral, it is inherently nonnegative, and can be zero only if $L_+\psi_l^m$ is identically zero. The right-hand side of Eq. (16.23), however, will become negative if m is permitted to get too large, so for any fixed l (and therefore a fixed λ_l), there must be some largest m , which we call $m_{\max}$, for which there exists a $\psi_l^{m_{\max}}$. But if we use Eq. (16.23) to evaluate $L_+\psi_l^{m_{\max}}$, we will, unless $\lambda_l - m_{\max}(m_{\max} + 1) = 0$, generate a function with $m = m_{\max} + 1$, thereby creating an inconsistency. Giving $m_{\max}$ the name l (permitted because within the current derivation we have not yet assigned a meaning to l), what we have found so far is that $\lambda_l = l(l + 1)$ and that the maximum value of m is $m = l$. Remember that we still know nothing about possible values for l .

Turning now to Eq. (16.24), and inserting $l(l + 1)$ for λ_l , we note that if m is permitted to become too negative we will again have an inconsistent situation, and that it is necessary that for some $m_{\min}$ the right-hand side of Eq. (16.24) must vanish. Thus, we require $l(l + 1) - m_{\min}(m_{\min} - 1) = 0$, an equation that is satisfied for $m_{\min} = l + 1$ (which is clearly irrelevant), and for $m_{\min} = -l$ (the solution we want).

Finally, we observe that, starting from some ψ_l^m with $m = l$, we have the severe limitation that application of the lowering operator L_- will decrease the m value in unit steps, but must ultimately reach $m = -l$ to avoid the generation of an inconsistency. This state of affairs is possible if l is a nonnegative integer, in which case there are $2l + 1$ possible m values, ranging in unit steps from l to $-l$. However, it is also possible to assign l a half-integer value, as $m = l$ and $m = -l$ are then still connected by a series of unit steps. In this case, also, there will be $2l + 1$ different m values. This quantity, $2l + 1$, is sometimes called the **multiplicity** of the angular momentum states.

The fact that it is mathematically possible to have a series (**multiplet**) of states corresponding to either integral or half-integral l and satisfying the angular momentum commutation rules does not prove that such states are realizable in a particular algebraic system (such as that describing ordinary three-dimensional [3-D] space), or that such states are indeed relevant for physics. However, by solving Laplace's equation, we have already found that the angular momentum states of integral l can be described in ordinary space and that they can be identified as states of ordinary (so-called **orbital**) angular momentum. It is not possible to describe states of half-integral l as ordinary functions in 3-D space, so orbital angular momentum will only involve integral l .

Example 16.1.1 SPHERICAL HARMONICS LADDER

From Exercise 3.10.30, or alternatively by combining the formulas for L_x and L_y from Eq. (16.13), the orbital angular momentum ladder operator L_+ is found to be

$$L_+ = e^{i\varphi} \left(\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \varphi} \right).$$

Starting from $Y_1^0(\theta, \varphi) = \sqrt{3/4\pi} \cos \theta$, we can apply Eq. (16.25):

$$L_+ Y_1^0(\theta, \varphi) = \sqrt{\frac{3}{4\pi}} e^{i\varphi} \frac{\partial \cos \theta}{\partial \theta} = \sqrt{\frac{3}{4\pi}} e^{i\varphi} (-\sin \theta) = \sqrt{2} Y_1^1(\theta, \varphi), \quad (16.26)$$

which when solved for Y_1^1 gives (with proper scale and sign) the value tabulated in Table 15.4. The reader can verify that the application of L_+ to Y_1^1 gives zero. ■

Spinors

It turns out that half-integral angular momentum states are needed to describe the intrinsic angular momentum of the electron and many other particles. Since these particles also have magnetic moments, an intuitive interpretation is that their charge distributions are spinning about some axis; hence the term **spin**. It is now understood that the spin phenomena cannot be explained consistently by describing these particles as ordinary charge distributions undergoing rotational motion, but are better treated by assigning these particles to states in an abstract space that, for the electron, has the l value $\frac{1}{2}$ (but in this context, we normally use s and write $s = \frac{1}{2}$), which means that the possible m values (often written m_s) are $m_s = +\frac{1}{2}$ and $m_s = -\frac{1}{2}$. It is not productive to try to think of this situation in terms of

ordinary functions, but to accept an abstract formulation in which spin states are represented by symbols; popular choices are α or $|\uparrow\rangle$ for the state $m = +\frac{1}{2}$ and β or $|\downarrow\rangle$ for that with $m = -\frac{1}{2}$. These spin states can also be represented by two-component column vectors, with the angular momentum operators given in terms of the Pauli matrices as $\frac{1}{2}\sigma_i$.

The quantities forming a basis for the multiplets for half-integer angular momentum are called **spinors**. In addition to their manipulation using ladder operators, they have rotational properties that are discussed in more detail in Chapter 17.

Example 16.1.2 SPINOR LADDER

Calling the angular momentum operator $\mathbf{S}$, we write S_x, S_y, S_z as the 2×2 matrices $\frac{1}{2}\sigma_i$, where σ_i are defined in Eq. (2.28):

$$S_x = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad S_y = \frac{1}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad S_z = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (16.27)$$

By carrying out matrix operations we can verify that these matrices satisfy the angular momentum commutation rules. For example,

$$\begin{aligned} S_x S_y - S_y S_x &= \frac{1}{4} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \frac{1}{4} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \frac{i}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = i S_z. \end{aligned}$$

We also find that

$$S_x^2 = S_y^2 = S_z^2 = (1/4)\mathbf{1},$$

and we therefore have

$$\mathbf{S}^2 = S_x^2 + S_y^2 + S_z^2 = \frac{1}{4}[\mathbf{1} + \mathbf{1} + \mathbf{1}] = \frac{3}{4}\mathbf{1}.$$

Note that $3/4$ is $S(S+1)$ for $S = 1/2$.

The interpretation of these matrix relationships is that we have an abstract space spanned by the two functions

$$\psi_{1/2}^{1/2} \equiv \alpha \equiv |\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \psi_{1/2}^{-1/2} \equiv \beta \equiv |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

and that the operators $\mathbf{S}^2$ and S_z operate on these functions as follows:

$$\mathbf{S}^2\alpha = \mathbf{S}^2\psi_{1/2}^{1/2} = \frac{3}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{3}{4} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{3}{4}\psi_{1/2}^{1/2} = \frac{3}{4}\alpha,$$

$$S_z\alpha = S_z\psi_{1/2}^{1/2} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{2}\psi_{1/2}^{1/2} = \frac{1}{2}\alpha,$$

$$\mathbf{S}^2\beta = \mathbf{S}^2\psi_{1/2}^{-1/2} = \frac{3}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{3}{4} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{3}{4}\psi_{1/2}^{-1/2} = \frac{3}{4}\beta,$$

$$S_z\beta = S_z\psi_{1/2}^{-1/2} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\frac{1}{2}\psi_{1/2}^{-1/2} = -\frac{1}{2}\beta.$$

The above formulas show that $\psi_{1/2}^{\pm 1/2}$ (also denoted α and β) are simultaneous eigenfunctions of $\mathbf{S}^2$ and S_z . To illustrate that they are not also eigenfunctions of S_x or S_y , we compute

$$S_x\alpha = S_x\psi_{1/2}^{1/2} = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{2}\psi_{1/2}^{-1/2} = \frac{1}{2}\beta.$$

To make ladders, we now form

$$S_+ = S_x + iS_y = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad S_- = S_x - iS_y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Applying these operators to $\alpha = \psi_{1/2}^{1/2}$,

$$S_+\alpha = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0, \quad S_-\alpha = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \beta.$$

These results are in agreement with Eq. (16.25), for which, with the current parameters $\lambda = 3/4$, $m = 1/2$, its coefficients are

$$\sqrt{\lambda - m(m+1)} = 0, \quad \sqrt{\lambda - m(m-1)} = 1.$$

■

Summary, Angular Momentum Formulas

The analysis of the preceding subsection applies to any system of operators satisfying the angular momentum commutation rules. Possible areas of application include orbital angular momentum (for which the eigenfunctions are the spherical harmonics), the intrinsic (spin) angular momentum we now know is associated with most fundamental particles, and even the overall angular momenta that result either from considering both the orbital and spin angular momenta of the same particle, or the total angular momentum of a collection of particles (as in a many-electron atom or even a nucleus).

It is useful to summarize the key results; we do so giving the operators the name J , to emphasize the fact that the results are not restricted to orbital angular momentum (for which the symbol L is nearly universally used), or to spin angular momentum (traditionally denoted S). Thus:

1. We assume that there exists a Hermitian operator $\mathbf{J}$ with components J_x, J_y, J_z such that $J_x^2 + J_y^2 + J_z^2 = \mathbf{J}^2$ and that these quantities satisfy the commutation relations

$$[J_k, J_l] = i\epsilon_{klm} J_m, \quad [\mathbf{J}^2, J_k] = 0, \quad (16.28)$$

where k, l, m are x, y, z in any order and ϵ_{klm} is a Levi-Civita symbol. Other than the requirement of Eq. (16.28), $\mathbf{J}$ is arbitrary.

2. Because the operators $\mathbf{J}^2$ and J_z commute, there can exist functions (in some abstract space), generically denoted ψ_J^M , with ψ_J^M simultaneously a normalized eigenfunction of J_z with eigenvalue M and an eigenfunction of $\mathbf{J}^2$ with eigenvalue $J(J+1)$:

$$J_z \psi_J^M = M \psi_J^M, \quad \mathbf{J}^2 \psi_J^M = J(J+1) \psi_J^M, \quad \langle \psi_J^M | \psi_J^M \rangle = 1. \quad (16.29)$$

3. Operators satisfying the above conditions can be called angular momentum operators; those which were used as examples of angular momentum in ordinary space (**orbital angular momentum**) are clearly relevant for physics; similar operators in more abstract spaces are relevant only to the extent that they can be identified with physical phenomena.

We have already seen that these assumptions are sufficient to enable the introduction of ladder operators, and to reach the following conclusions:

1. The possible values of J are integral and half-integral; in ordinary 3-D space only functions of integral J can be realized.
2. For a given J , the possible values of M range in unit steps from $M = J$ to $M = -J$; this produces $2J+1$ different M values.
3. Given any one ψ_J^M , we can generate others by use of the operators

$$J_+ = J_x + iJ_y, \quad J_- = J_x - iJ_y.$$

The result of applying these operators to ψ_J^M is, see Eq. (16.25),

$$J_+ \psi_J^M = \sqrt{(J-M)(J+M+1)} \psi_J^{M+1}, \quad (16.30)$$

$$J_- \psi_J^M = \sqrt{(J+M)(J-M+1)} \psi_J^{M-1}. \quad (16.31)$$

These formulas give zero results when J_+ is applied to ψ_J^J and when J_- is applied to ψ_J^{-J} .

Exercises

- 16.1.1** The quantum mechanical angular momentum operators $L_x \pm iL_y$ in 3-D physical space are given by

$$L_x + iL_y = e^{i\varphi} \left(\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \varphi} \right),$$

$$L_x - iL_y = -e^{-i\varphi} \left(\frac{\partial}{\partial \theta} - i \cot \theta \frac{\partial}{\partial \varphi} \right).$$

Show that

$$(a) \quad (L_x + iL_y)Y_L^M(\theta, \varphi) = \sqrt{(L-M)(L+M+1)}Y_L^{M+1}(\theta, \varphi),$$

$$(b) \quad (L_x - iL_y)Y_L^M(\theta, \varphi) = \sqrt{(L+M)(L-M+1)}Y_L^{M-1}(\theta, \varphi).$$

16.1.2 With $L_{\pm}$ given by

$$L_{\pm} = L_x \pm iL_y = \pm e^{\pm i\varphi} \left[\frac{\partial}{\partial \theta} \pm i \cot \theta \frac{\partial}{\partial \varphi} \right],$$

show that

$$(a) \quad Y_l^m = \sqrt{\frac{(l+m)!}{(2l)!(l-m)!}}(L_-)^{l-m}Y_l^l,$$

$$(b) \quad Y_l^m = \sqrt{\frac{(l-m)!}{(2l)!(l+m)!}}(L_+)^{l+m}Y_l^{-l}.$$

16.1.3 Using the known forms of L_+ and L_- (Exercise 16.1.2), show that

$$\int [Y_L^M]^* L_- (L_+ Y_L^M) d\Omega = \int (L_+ Y_L^M)^* (L_+ Y_L^M) d\Omega.$$

Here $d\Omega$ is the element of solid angle ($\sin \theta d\theta d\varphi$), and the integration is over the entire angular space.

- 16.1.4** (a) Show that $\mathbf{J}^2 = \frac{1}{2} [J_+ J_- + J_- J_+] + J_z^2$.
 (b) Use the result from part (a) and the explicit formulas for L_+ and L_- from Exercise 16.1.2 to verify that all the spherical harmonics with $l = 2$ are eigenfunctions of $\mathbf{L}^2$ with eigenvalue $l(l+1) = 6$.

16.1.5 Derive the following relations without assuming anything about ψ_L^M other than that they are angular momentum eigenfunctions:

$$(a) \quad \psi_L^M(\theta, \varphi) = \sqrt{\frac{(L+M)!}{(2L)!(L-M)!}}(L_-)^{L-M}\psi_L^L(\theta, \varphi),$$

$$(b) \quad \psi_L^M(\theta, \varphi) = \sqrt{\frac{(L-M)!}{(2L)!(L+M)!}}(L_+)^{L+M}\psi_L^{-L}(\theta, \varphi).$$

16.1.6 Derive the operator equations

$$(L_+)^n Y_L^M(\theta, \varphi) = (-1)^n e^{in\varphi} \sin^{n+M} \theta \frac{d^n \sin^{-M} \theta Y_L^M(\theta, \varphi)}{(d \cos \theta)^n},$$

$$(L_-)^n Y_L^M(\theta, \varphi) = e^{-in\varphi} \sin^{n-M} \theta \frac{d^n \sin^M \theta Y_L^M(\theta, \varphi)}{(d \cos \theta)^n}.$$

Hint. Try mathematical induction (Section 1.4).

16.1.7 Show, using $(L_-)^n$, that

$$Y_L^{-M}(\theta, \varphi) = (-1)^M \left[Y_L^M(\theta, \varphi) \right]^*.$$

16.1.8 Verify by explicit calculation that

$$(a) \quad L_+ Y_1^0(\theta, \varphi) = -\sqrt{\frac{3}{4\pi}} \sin \theta e^{i\varphi} = \sqrt{2} Y_1^1(\theta, \varphi),$$

$$(b) \quad L_- Y_1^0(\theta, \varphi) = +\sqrt{\frac{3}{4\pi}} \sin \theta e^{-i\varphi} = \sqrt{2} Y_1^{-1}(\theta, \varphi).$$

The signs have the indicated values because the spherical harmonics were defined to be consistent with the results obtained using the ladder operators L_+ and L_- (Condon-Shortley phase).

16.2 ANGULAR MOMENTUM COUPLING

An important application of ladder operators is to systems in which a resultant angular momentum is the sum of two individual angular momenta. Because the angular momenta have directional properties, we anticipate a result that has some properties in common with vector addition, but because these are quantum mechanical quantities involving non-commuting operators, we need to study the problem in more detail.

If $\mathbf{j}_1$ and $\mathbf{j}_2$ are two individual angular momentum operators that act on different coordinate sets (as, e.g., the coordinates of two different particles), then they are unrelated and all components of each must commute with every component of the other. This will enable us to carry out a detailed analysis of operators of the combined system, for which the total angular momentum operator is $\mathbf{J} = \mathbf{j}_1 + \mathbf{j}_2$, with components $J_x = j_{1x} + j_{2x}$, $J_y = j_{1y} + j_{2y}$, $J_z = j_{1z} + j_{2z}$, with the overall operator $\mathbf{J}^2 = J_x^2 + J_y^2 + J_z^2$.

To discuss the problem, we will need the commutators

$$[j_{1k}, j_{1l}] = i\epsilon_{klm} j_{1m}, \quad [j_{2k}, j_{2l}] = i\epsilon_{klm} j_{2m}, \quad [j_{1k}, j_{2l}] = 0. \quad (16.32)$$

For the first two commutators, k, l, m are x, y, z in any order; the third commutator vanishes for all k, l including $k = l$. From the commutators in Eq. (16.32), it is easily established that the overall angular momentum components obey the commutation rules

$$[J_x, J_y] = iJ_z, \quad [J_y, J_z] = iJ_x, \quad [J_z, J_x] = iJ_y, \quad (16.33)$$

so these overall components satisfy the generic angular momentum commutation relations, meaning also that

$$[\mathbf{J}^2, J_i] = 0. \quad (16.34)$$

In addition,

$$[\mathbf{J}^2, \mathbf{j}_1^2] = [\mathbf{J}^2, \mathbf{j}_2^2] = 0. \quad (16.35)$$

However, it is **not** true that the components of $\mathbf{j}_1$ or $\mathbf{j}_2$, namely j_{1i} or j_{2i} , commute with $\mathbf{J}^2$, even though $\mathbf{J}^2$ and the sum $j_{1i} + j_{2i}$ do commute.

Example 16.2.1 COMMUTATION RULES FOR $\mathbf{J}$ COMPONENTS

To find the commutator $[\mathbf{J}^2, j_{1z}]$, write

$$\begin{aligned} \mathbf{J}^2 &= (\mathbf{j}_1 + \mathbf{j}_2)^2 = \mathbf{j}_1^2 + \mathbf{j}_2^2 + 2\mathbf{j}_1 \cdot \mathbf{j}_2 \\ &= \mathbf{j}_1^2 + \mathbf{j}_2^2 + 2(j_{1x}j_{2x} + j_{1y}j_{2y} + j_{1z}j_{2z}), \end{aligned}$$

so we have

$$\begin{aligned} [\mathbf{J}^2, j_{1z}] &= [\mathbf{j}_1^2, j_{1z}] + [\mathbf{j}_2^2, j_{1z}] + 2([j_{1x}j_{2x}, j_{1z}] + [j_{1y}j_{2y}, j_{1z}] + [j_{1z}j_{2z}, j_{1z}]) \\ &= 2(j_{2x}[j_{1x}, j_{1z}] + j_{2y}[j_{1y}, j_{1z}]) = 2i(j_{1x}j_{2y} - j_{1y}j_{2x}), \end{aligned} \quad (16.36)$$

where we have dropped terms in which the commutators involve different particles and those, e.g., $[\mathbf{j}_1^2, j_{1z}]$, which vanish because the individual-particle operators are angular momenta.

Equation (16.36) clearly shows that $[\mathbf{J}^2, j_{1z}]$ is nonzero. However, its contributions are equal and opposite to those of $[\mathbf{J}^2, j_{2z}]$, explaining why $[\mathbf{J}^2, J_z]$ does vanish.

Consider next $[\mathbf{J}^2, \mathbf{j}_1^2]$. Again expanding $\mathbf{J}^2$, we get

$$[\mathbf{J}^2, \mathbf{j}_1^2] = [\mathbf{j}_1^2, \mathbf{j}_1^2] + [\mathbf{j}_2^2, \mathbf{j}_1^2] + 2([j_{1x}j_{2x}, \mathbf{j}_1^2] + [j_{1y}j_{2y}, \mathbf{j}_1^2] + [j_{1z}j_{2z}, \mathbf{j}_1^2]).$$

Every term of this equation vanishes, so $\mathbf{J}^2$ and $\mathbf{j}_1^2$ commute. ■

We have noted that $\mathbf{j}_1^2$, $\mathbf{j}_2^2$, and J_z all commute with each other and with $\mathbf{J}^2$, j_{1z} , and j_{2z} , but that the last three of these operators do not all commute with each other. There are therefore different ways of selecting maximal sets of mutually commuting operators for which we can construct simultaneous eigenfunctions. One possibility is to select $\mathbf{j}_1^2$, $\mathbf{j}_2^2$, j_{1z} , j_{2z} , and J_z , which has the advantage that the simultaneous eigenfunctions are just products of the eigenstates for individual $\mathbf{j}_i$, but has the disadvantage that we will not have states of definite total angular momentum $\mathbf{J}^2$. This is a big disadvantage, because in reality different angular momenta in the same system actually interact to some extent. If we add to the Hamiltonian of our problem a small term (a **perturbation**) that causes the individual angular momenta not quite to be independent, our system will still strictly have conservation of $\mathbf{J}^2$ (i.e., H and $\mathbf{J}^2$ will still commute), but the perturbation added to the Hamiltonian will not commute with $\mathbf{j}_1$ and $\mathbf{j}_2$.

Alternatively, and for most purposes better, we could choose the mutually commuting operator set $\mathbf{J}^2$, $\mathbf{j}_1^2$, $\mathbf{j}_2^2$, and J_z , which would describe states of definite total angular momentum, but these states would be mixtures of the individual angular-momentum states and would not have definite values of j_{1z} or j_{2z} . It is the purpose of this section to relate these two descriptions by finding the equations that connect (i.e., **couple**) the individual angular momenta to form states of definite $\mathbf{J}^2$.

To simplify future discussion, we can refer to the product basis of the preceding paragraph as the m_1, m_2 basis, and call the alternative basis of definite J the J, M basis. The m_1, m_2 basis members also have definite values of M , but not J ; most members of the J, M basis will not have definite values of either m_1 or m_2 . If we stick with problems in which j_1 and j_2 are fixed, all members of both bases will have the same definite values of these quantum numbers.

Before getting into the details, let's make two observations. First, since we have raising and lowering operators that we can apply to the J, M basis, the functions in this basis must include all the M values for any J that is present at all. Second, since both bases have definite values of M , the transition from one basis to the other cannot mix functions of different M .

Vector Model

We begin with some qualitative observations. Since $J_z = j_{1z} + j_{2z}$ (with eigenvalues we call M) is part of both our commuting operator sets, we can conclude, from looking at the m_1, m_2 basis, that the maximum eigenvalue $M_{\max}$ of J_z will occur when $m_1 = j_1$ and $m_2 = j_2$, so $M_{\max} = j_1 + j_2$. Moving now to the J, M basis, which of course spans the same function space, we see that because $M_{\max}$ is the maximum M value, it must be a member of a multiplet with $J = M_{\max}$, and this must be the largest possible J . Thus, $J_{\max} = j_1 + j_2$.

To establish the minimum value possible for the quantum number J is a little trickier, and we will come back to that shortly. The result, which is simple, is that $J_{\min} = |j_1 - j_2|$. These maximum and minimum values of J correspond to the notion that the classical vector sum $\mathbf{j}_1 + \mathbf{j}_2$ has a maximum length equal to the sum of the lengths of these vectors and a minimum length equal to the absolute value of their difference; the quantum analog of this notion is not quantitatively exact because the magnitude of each $\mathbf{j}$ is actually $\sqrt{j(j+1)}$.

Further qualitative observations follow if we tabulate the various possible m_1, m_2 functions of various M values. The concept can be understood from a simple example. Suppose $j_1 = 2$, $j_2 = 1$. Then the members of the m_1, m_2 basis can be grouped as shown here. The kets in the table are labeled in more detail than usual to avoid potential confusion; those labeled m_1 have j value j_1 , those labeled m_2 have $j = j_2$.

$$\begin{array}{l}
 M = +3 \quad |m_1 = +2\rangle|m_2 = +1\rangle \\
 M = +2 \quad |m_1 = +2\rangle|m_2 = 0\rangle \quad |m_1 = +1\rangle|m_2 = +1\rangle \\
 M = +1 \quad |m_1 = +2\rangle|m_2 = -1\rangle \quad |m_1 = +1\rangle|m_2 = 0\rangle \quad |m_1 = 0\rangle|m_2 = +1\rangle \\
 M = 0 \quad |m_1 = +1\rangle|m_2 = -1\rangle \quad |m_1 = 0\rangle|m_2 = 0\rangle \quad |m_1 = -1\rangle|m_2 = +1\rangle \\
 M = -1 \quad |m_1 = -2\rangle|m_2 = +1\rangle \quad |m_1 = -1\rangle|m_2 = 0\rangle \quad |m_1 = 0\rangle|m_2 = -1\rangle \\
 M = -2 \quad |m_1 = -2\rangle|m_2 = 0\rangle \quad |m_1 = -1\rangle|m_2 = -1\rangle \\
 M = -3 \quad |m_1 = -2\rangle|m_2 = -1\rangle
 \end{array}$$

Because the basis transformations we are discussing only mix basis functions of the same M , a transition to the J, M basis will have the same number of functions of each M as are in the row of our table for that M . If there is only one function in the row, it must (without change) be a member of the J, M basis, and we may use it as a starting point for getting all the other members of the multiplet for the same J by application of the ladder operators. So in our current example, we can start from $|m_1 = +2\rangle|m_2 = +1\rangle$, and make one member of the multiplet for each M value in the table.

Once this has been done, we will have constructed as many J, M functions as there are entries in the first column of the table (but remember that in most cases they will not be the specific functions sitting in that column). But that observation does tell us that the numbers of functions that are still unused (but not their exact forms) will correspond with the numbers of functions in the remainder of the table. In particular, we see that there will in our example be one function left over with $M = 2$. Because it cannot have a $J = 3$ component, it must be a $|J = 2, M = 2\rangle$ eigenfunction and therefore must be orthogonal to the $|J = 3, M = 2\rangle$ function we have already found. That means that we can obtain it by Gram-Schmidt orthogonalization within the function space for $M = +2$.

From the $|J = 2, M = 2\rangle$ function, we can apply a ladder operator to find $|J = 2, M\rangle$ basis members with other M values, the number of which will correspond to the number of entries in the second column of our table. To continue to a third column, we would need to find a $M = +1$ function orthogonal to **both** the $|J = 3, M = +1\rangle$ and $|J = 2, M = +1\rangle$ functions. This process can be continued until the m_1, m_2 basis has been exhausted.

Taking now a further look at our table, we see that the number of columns with entries increases as we decrease M from its maximum value until M has reached $|j_1 - j_2|$; for smaller $|M|$ than that, the number of columns in use stays constant, because of limitations in the way the individual m values can be chosen to add up to M . That gives us a graphical indication that the smallest J value will be $|j_1 - j_2|$.

A more algebraic way of determining the smallest resultant J is based on a computation of the total number of J, M states generated if the possible J values run from an as yet undetermined value $J_{\min}$ to our previously determined maximum value $J_{\max}$. Since the number of states for each J is $2J + 1$, the total number of J, M states we will have produced is

$$\begin{aligned} \sum_{J=J_{\min}}^{J=J_{\max}} (2J + 1) &= (J_{\max} - J_{\min} + 1)(J_{\max} + J_{\min} + 1) \\ &= (2j_1 + 1)(2j_2 + 1), \end{aligned} \tag{16.37}$$

where the second line of this equation reflects the fact that the total number of states is readily counted in the m_1, m_2 basis. Inserting the value $J_{\max} = j_1 + j_2$ and solving for $J_{\min}$, we find

$$J_{\min} = |j_1 - j_2|.$$

Another way of stating this result is to observe that the possible values of J satisfy a **triangle rule**, meaning that they occur in unit steps from a maximum of $j_1 + j_2$ to a minimum of $|j_1 - j_2|$.

Ladder Operator Construction

To develop a quantitative description of angular momentum coupling, we consider the case of general j_1 and j_2 , and start from the lone member of the m_1, m_2 basis with $M = j_1 + j_2$. In line with our earlier discussion, this m_1, m_2 basis member must also be a function of the definite J value $J_{\max} = j_1 + j_2$. Using a notation in which the lower entry in each ket is its J value and the upper entry gives the value of M , we indicate this by writing

$$\begin{vmatrix} J_{\max} \\ J_{\max} \end{vmatrix} = \begin{vmatrix} j_1 \\ j_1 \end{vmatrix} \begin{vmatrix} j_2 \\ j_2 \end{vmatrix}. \quad (16.38)$$

We now generate additional states of the same J but with different M by applying the lowering operator J_- to Eq. (16.38); when we apply it to the right-hand side, we do so in the form $J_- = j_{1-} + j_{2-}$. The result, for the left side of Eq. (16.38), is

$$J_- \begin{vmatrix} J_{\max} \\ J_{\max} \end{vmatrix} = \sqrt{2J_{\max}} \begin{vmatrix} J_{\max} - 1 \\ J_{\max} \end{vmatrix}. \quad (16.39)$$

The coefficient $\sqrt{2J_{\max}}$ is that given by Eq. (16.31) for $J = M = J_{\max}$. For the right side of Eq. (16.38), we get

$$\begin{aligned} (j_{1-} + j_{2-}) \left[\begin{vmatrix} j_1 \\ j_1 \end{vmatrix} \begin{vmatrix} j_2 \\ j_2 \end{vmatrix} \right] &= \left[j_{1-} \begin{vmatrix} j_1 \\ j_1 \end{vmatrix} \right] \begin{vmatrix} j_2 \\ j_2 \end{vmatrix} + \begin{vmatrix} j_1 \\ j_1 \end{vmatrix} \left[j_{2-} \begin{vmatrix} j_2 \\ j_2 \end{vmatrix} \right] \\ &= \sqrt{2j_1} \begin{vmatrix} j_1 - 1 \\ j_1 \end{vmatrix} \begin{vmatrix} j_2 \\ j_2 \end{vmatrix} + \sqrt{2j_2} \begin{vmatrix} j_1 \\ j_1 \end{vmatrix} \begin{vmatrix} j_2 - 1 \\ j_2 \end{vmatrix}, \end{aligned} \quad (16.40)$$

where we have again obtained the coefficients from Eq. (16.31), but now evaluating them for the first term with $(J, M) = (j_1, j_1)$ and for the second term with $(J, M) = (j_2, j_2)$.

Combining these results, and solving for $\begin{vmatrix} J_{\max} - 1 \\ J_{\max} \end{vmatrix}$,

$$\begin{vmatrix} J_{\max} - 1 \\ J_{\max} \end{vmatrix} = \sqrt{\frac{j_1}{J_{\max}}} \begin{vmatrix} j_1 - 1 \\ j_1 \end{vmatrix} \begin{vmatrix} j_2 \\ j_2 \end{vmatrix} + \sqrt{\frac{j_2}{J_{\max}}} \begin{vmatrix} j_1 \\ j_1 \end{vmatrix} \begin{vmatrix} j_2 - 1 \\ j_2 \end{vmatrix}. \quad (16.41)$$

With escalating complexity, we could continue this process to smaller values of M .

As indicated in our earlier, more qualitative discussion, we can reach functions with $J = J_{\max} - 1$ by starting from the unused member of the set of two functions

$$\begin{vmatrix} j_1 - 1 \\ j_1 \end{vmatrix} \begin{vmatrix} j_2 \\ j_2 \end{vmatrix}, \quad \begin{vmatrix} j_1 \\ j_1 \end{vmatrix} \begin{vmatrix} j_2 - 1 \\ j_2 \end{vmatrix}.$$

The quantity we seek,

$$\begin{vmatrix} J_{\max} - 1 \\ J_{\max} - 1 \end{vmatrix},$$

will be the function in the above-defined subspace that is orthogonal to

$$\left| \begin{matrix} J_{\max} - 1 \\ J_{\max} \end{matrix} \right\rangle$$

as given in Eq. (16.41), and therefore will be

$$\left| \begin{matrix} J_{\max} - 1 \\ J_{\max} \end{matrix} \right\rangle = -\sqrt{\frac{j_2}{J_{\max}}} \left| \begin{matrix} j_1 - 1 \\ j_1 \end{matrix} \right\rangle \left| \begin{matrix} j_2 \\ j_2 \end{matrix} \right\rangle + \sqrt{\frac{j_1}{J_{\max}}} \left| \begin{matrix} j_1 \\ j_1 \end{matrix} \right\rangle \left| \begin{matrix} j_2 - 1 \\ j_2 \end{matrix} \right\rangle. \quad (16.42)$$

At this point we note that the function produced by Eq. (16.42) could have been written with all its signs changed, as the orthogonalization process does not determine the sign of the orthogonal function. This only matters if we wish to correlate the signs of our J, M constructions with work by others. Irrespective of our choice of signs, we can apply J_- to reach the full set of M values, and then continue to states of smaller J until the m_1, m_2 space is exhausted.

The general result of the above-described processes is to obtain each J, M eigenstate as a linear combination of m_1, m_2 states of the same M , in a fashion summarized by the following equation (written in a less cumbersome notation now that the need for detail has disappeared):

$$|J, M\rangle = \sum_{m_1, m_2} C(j_1, j_2, J | m_1, m_2, M) |j_1, m_1; j_2, m_2\rangle. \quad (16.43)$$

Here $|j_1, m_1; j_2, m_2\rangle$ stands for $|j_1, m_1\rangle |j_2, m_2\rangle$ and we have given over to the coefficient $C(j_1, j_2, J | m_1, m_2, M)$ the responsibility to vanish when $m_1 + m_2 \neq M$. Thus, the apparent double summation in Eq. (16.43) is actually a single sum. The coefficients in Eq. (16.43) are called **Clebsch-Gordan coefficients**. To resolve the sign ambiguity resulting from the orthogonalization processes, they are defined to have signs specified by the Condon-Shortley phase convention.

It is important to realize that all the results of this section remain valid irrespective of whether j_1, j_2 , or both are integral or half-integral. For example, if $j_1 = 1$ and $j_2 = \frac{1}{2}$ (corresponding to the coupling of the orbital and spin angular momenta of an electron), the possible J, M states will be a **quartet** for $J = 3/2$ (with M values $+3/2, +1/2, -1/2, -3/2$), and a **doublet** for $J = 1/2$ (with M values $+1/2$ and $-1/2$).

A second way to look at the Clebsch-Gordan coefficients is to identify them as the scalar products

$$C(j_1, j_2, J | m_1, m_2, M) = \langle J, M | j_1, m_1; j_2, m_2 \rangle. \quad (16.44)$$

Because of the method used for the construction of the $|J, M\rangle$, we can make one additional observation: The Clebsch-Gordan coefficients will all be real, even if the $|j_1, m_1\rangle$ and $|j_2, m_2\rangle$ used for their construction are not.

The Clebsch-Gordan expansion can be interpreted in yet another way. We can view the Clebsch-Gordan coefficients as the elements of a transformation matrix converting functions of the m_1, m_2 basis into those of the J, M basis; since both basis sets are orthonormal, the transformation must be unitary (and because it is real, orthogonal). This means that the inverse transformation, $(J, M) \rightarrow (m_1, m_2)$, must have a transformation matrix that is the

transpose of that for the forward transformation $(m_1, m_2) \rightarrow (J, M)$. That means that we also have the equation

$$|j_1, m_1; j_2, m_2\rangle = \sum_{JM} C(j_1, j_2, J|m_1, m_2, M)|J, M\rangle. \quad (16.45)$$

This equation is correct and corresponds to our discussion. Note that instead of reversing the index order of the transformation matrix we have interchanged the index sets identifying the functions.

In passing, we make one further comment. While the Clebsch-Gordan coefficients can be identified as forming a transformation matrix, note that their row/column indexing differs from the pattern to which we are accustomed, since, instead of labels running from 1 to n (the dimension of the transformation), we are using in one dimension the compound index (m_1, m_2) , and in the other dimension the compound quantity (J, M) . This Clebsch-Gordan matrix will be somewhat sparse (containing many zero elements). The zeros occur because the coefficients vanish unless $M = m_1 + m_2$.

There is a significant literature on the practical computation of Clebsch-Gordan coefficients,¹ but to make the present discussion complete we simply give here a closed general formula:

$$C(j_1, j_2, J|m_1, m_2, M) = F_1 F_2 F_3, \quad (16.46)$$

where

$$\begin{aligned} F_1 &= \sqrt{\frac{(j_1 + j_2 - J)!(J + j_1 - j_2)!(J + j_2 - j_1)!(2J + 1)}{(j_1 + j_2 + J + 1)!}} \\ F_2 &= \sqrt{(J + M)!(J - M)!(j_1 + m_1)!(j_1 - m_1)!(j_2 + m_2)!(j_2 - m_2)!}, \\ F_3 &= \sum_s \frac{(-1)^s}{(j_1 - m_1 - s)!(j_2 + m_2 - s)!(J - j_2 + m_1 + s)!} \\ &\quad \times \frac{1}{(J - j_1 - m_2 + s)!(j_1 + j_2 - J - s)!s!}. \end{aligned}$$

The F_3 summation is over all integer values of s for which the factorials all have non-negative arguments (which will be integral). The sum is therefore finite in extent and F_3 is a closed form. Equation (16.46) is only to be used for parameter values that satisfy the angular momentum and coupling conditions: j_1, j_2, J must satisfy the triangle condition, m_i is to be from the sequence $l_i, l_i - 1, \dots, -l_i$ ($i = 1, 2$), M to be from $J, J - 1, \dots, -J$, and $M = m_1 + m_2$.

Finally, we call attention to the fact that Clebsch-Gordan coefficients have symmetries that are not obvious from the foregoing development. To expose the symmetries, it is convenient to convert them to the Wigner 3- j -symbols, defined as

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = \frac{(-1)^{j_1 - j_2 - m_3}}{(2j_3 + 1)^{1/2}} C(j_1, j_2, j_3|m_1, m_2, -m_3). \quad (16.47)$$

¹See Biedenharn and Louck, Brink and Satchler, Edmonds, Rose, and Wigner in Additional Readings. Clebsch-Gordan coefficients are also tabulated in many places, and can easily be found online by a Web search.

Extensive discussion of $3j$ -symbols and related quantities is beyond the scope of this book. This important, but advanced, topic is presented in most of the sources listed under Additional Readings.

The $3j$ -symbols are invariant under even permutations of the indices (1,2,3), but under odd permutations $(1, 2, 3) \rightarrow (k, l, n)$ transform as follows:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = (-1)^{j_1+j_2+j_3} \begin{pmatrix} j_k & j_l & j_n \\ m_k & m_l & m_n \end{pmatrix}. \quad (16.48)$$

They also have the following symmetry under change of sign of their lower indices:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = (-1)^{j_1+j_2+j_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ -m_1 & -m_2 & -m_3 \end{pmatrix}. \quad (16.49)$$

Even though some of the j_i may be half-integral, remember that j_3 must be equal to $j_1 + j_2$ or differ therefrom by an integer. This fact causes the powers of -1 in Eqs. (16.47) through (16.49) to be integral, so these factors are not multiple-valued and the sign assignments of the $3j$ -symbols are unambiguous. These symmetry relations make a table of $3j$ -symbols more compact than one of Clebsch-Gordan coefficients; such tables can be found in the literature,² and a short list is included here, as Table 16.1.

We close this section with two examples.

Table 16.1 Wigner $3j$ -Symbols

$\begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 1 \\ \frac{1}{2} & -\frac{1}{2} & 0 \end{pmatrix} = \frac{1}{\sqrt{6}}$	$\begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 1 \\ \frac{1}{2} & \frac{1}{2} & -1 \end{pmatrix} = -\frac{1}{\sqrt{3}}$	$\begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \end{pmatrix} = \frac{1}{\sqrt{2}}$
$\begin{pmatrix} 1 & \frac{1}{2} & -\frac{3}{2} \\ 1 & \frac{1}{2} & -\frac{3}{2} \end{pmatrix} = \frac{1}{2}$	$\begin{pmatrix} 1 & \frac{1}{2} & -\frac{3}{2} \\ 1 & -\frac{1}{2} & -\frac{1}{2} \end{pmatrix} = -\frac{1}{\sqrt{12}}$	$\begin{pmatrix} 1 & \frac{1}{2} & -\frac{3}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} \end{pmatrix} = -\frac{1}{\sqrt{6}}$
$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = -\frac{1}{\sqrt{3}}$	$\begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \end{pmatrix} = \frac{1}{\sqrt{3}}$	$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} = 0$
$\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \end{pmatrix} = \frac{1}{\sqrt{6}}$	$\begin{pmatrix} 1 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} = \sqrt{\frac{2}{15}}$	$\begin{pmatrix} 1 & 1 & 2 \\ 1 & -1 & 0 \end{pmatrix} = \frac{1}{\sqrt{30}}$
$\begin{pmatrix} 1 & 1 & 2 \\ 1 & 0 & -1 \end{pmatrix} = -\frac{1}{\sqrt{10}}$	$\begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & -2 \end{pmatrix} = \frac{1}{\sqrt{5}}$	$\begin{pmatrix} 1 & 2 & 2 \\ 1 & -1 & 0 \end{pmatrix} = -\frac{1}{\sqrt{10}}$
$\begin{pmatrix} 1 & 2 & 2 \\ 1 & -2 & 1 \end{pmatrix} = \frac{1}{\sqrt{15}}$	$\begin{pmatrix} 1 & 2 & 2 \\ 0 & 1 & -1 \end{pmatrix} = -\frac{1}{\sqrt{30}}$	$\begin{pmatrix} 1 & 2 & 2 \\ 0 & 2 & -2 \end{pmatrix} = \sqrt{\frac{2}{15}}$
$\begin{pmatrix} 1 & 2 & 2 \\ 0 & 0 & 0 \end{pmatrix} = 0$	$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{pmatrix} = -\sqrt{\frac{3}{35}}$	$\begin{pmatrix} 1 & 2 & 3 \\ 1 & -1 & 0 \end{pmatrix} = -\frac{1}{\sqrt{35}}$
$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & -1 \end{pmatrix} = \sqrt{\frac{2}{35}}$	$\begin{pmatrix} 1 & 2 & 3 \\ 1 & -2 & 1 \end{pmatrix} = \frac{1}{\sqrt{105}}$	$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & -2 \end{pmatrix} = -\sqrt{\frac{2}{21}}$
$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & -3 \end{pmatrix} = \frac{1}{\sqrt{7}}$	$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & -1 \end{pmatrix} = \sqrt{\frac{8}{105}}$	$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 2 & -2 \end{pmatrix} = -\frac{1}{\sqrt{21}}$

²See, for example, M. Rotenberg, R. Bivins, N. Metropolis, and J. K. Wooten, Jr., *The $3j$ - and $6j$ -Symbols*. Cambridge, MA: Massachusetts Institute of Technology Press (1959).

Example 16.2.2 TWO SPINORS

This example describes a problem that exists entirely in an abstract space, namely the coupling of two spin- $\frac{1}{2}$ particles (e.g., electrons) to form combined states of definite J . Letting α stand for a normalized single-particle state with $j = \frac{1}{2}$, $m = +\frac{1}{2}$, with β a normalized state with $j = \frac{1}{2}$, $m = -\frac{1}{2}$, we have the following four states in the m_1, m_2 basis:

$$\begin{aligned} M = 1: & \quad \alpha\alpha \\ M = 0: & \quad \alpha\beta \quad \beta\alpha \\ M = -1: & \quad \beta\beta \end{aligned}$$

For all these two-particle states, the first symbol refers to particle 1, the second to particle 2. From Eq. (16.31) and Example 16.1.2, we have $j_-\alpha = \beta$, $j_-\beta = 0$, and we can use the following rearrangement of Eq. (16.31) to deal with the $|J, M\rangle$ states. Again we use a notation in which the lower entry in the ket is J ; the upper entry is M :

$$\begin{vmatrix} M-1 \\ J \end{vmatrix} = \frac{1}{\sqrt{(J+M)(J-M+1)}} J_- \begin{vmatrix} M \\ J \end{vmatrix}. \quad (16.50)$$

The maximum M value in this system is $M = +1$, so the one state of this M value must have $J = 1$. showing that $\begin{vmatrix} 1 \\ 1 \end{vmatrix} = \alpha\alpha$. Starting from it, we lower M :

$$\begin{aligned} \begin{vmatrix} 1 \\ 1 \end{vmatrix} &= \alpha\alpha, \\ \begin{vmatrix} 0 \\ 1 \end{vmatrix} &= \frac{1}{\sqrt{2}} J_- \begin{vmatrix} 1 \\ 1 \end{vmatrix} = \frac{1}{\sqrt{2}} \beta\alpha + \frac{1}{\sqrt{2}} \alpha\beta, \\ \begin{vmatrix} -1 \\ 1 \end{vmatrix} &= \frac{1}{\sqrt{2}} J_- \begin{vmatrix} 0 \\ 1 \end{vmatrix} = \frac{1}{\sqrt{2}} \left[\frac{1}{\sqrt{2}} \beta\beta + \frac{1}{\sqrt{2}} \beta\beta \right] = \beta\beta. \end{aligned}$$

These are the well-known members of the $S = 1$ spin multiplet, which is known as a **triplet**. At $M = 0$, where there were two m_1, m_2 states, the state orthogonal to $\begin{vmatrix} 0 \\ 1 \end{vmatrix}$ must be the $\begin{vmatrix} 0 \\ 0 \end{vmatrix}$ state.

Even though we do not have an entirely explicit representation of the states α and β , we do know that they are normalized eigenstates of a Hermitian operator (J_z) with different eigenvalues, and therefore they must be orthogonal. Thus, we can apply the Gram-Schmidt process to the $M = 0$ subspace, using the relations

$$\langle \alpha | \alpha \rangle = \langle \beta | \beta \rangle = 1, \quad \langle \alpha | \beta \rangle = 0.$$

We easily find that the normalized function orthogonal to $(\alpha\beta + \beta\alpha)/\sqrt{2}$ is $(\alpha\beta - \beta\alpha)/\sqrt{2}$. It is the only member of the $S = 0$ multiplet, and is therefore known as a **singlet**. Note that we didn't have to know anything specific about spin operators to carry out this analysis.

Our tableau of states can now be written in the J, M basis:

$$\begin{array}{ccc}
 & J = 1 & J = 0 \\
 M = 1 & \alpha\alpha & \\
 M = 0 & (\alpha\beta + \beta\alpha)/\sqrt{2} & (\alpha\beta - \beta\alpha)/\sqrt{2} \\
 M = -1 & \beta\beta &
 \end{array}$$

From the J, M tableau, we can read out the Clebsch-Gordan coefficients:

$$\begin{aligned}
 C\left(\frac{1}{2}, \frac{1}{2}, 1 \mid \frac{1}{2}, \frac{1}{2}, 1\right) &= 1 \\
 C\left(\frac{1}{2}, \frac{1}{2}, 1 \mid \frac{1}{2}, -\frac{1}{2}, 0\right) &= C\left(\frac{1}{2}, \frac{1}{2}, 1 \mid -\frac{1}{2}, \frac{1}{2}, 0\right) = \frac{1}{\sqrt{2}} \\
 C\left(\frac{1}{2}, \frac{1}{2}, 0 \mid \frac{1}{2}, -\frac{1}{2}, 0\right) &= -C\left(\frac{1}{2}, \frac{1}{2}, 1 \mid -\frac{1}{2}, \frac{1}{2}, 0\right) = \frac{1}{\sqrt{2}} \\
 C\left(\frac{1}{2}, \frac{1}{2}, 1 \mid -\frac{1}{2}, -\frac{1}{2}, -1\right) &= 1
 \end{aligned}$$

These coefficients can also be obtained from our table of $3j$ -symbols. Using Eq. (16.47), we find the coefficients for $|J = 1, M = 0\rangle$ to be

$$\begin{aligned}
 C\left(\frac{1}{2}, \frac{1}{2}, 1 \mid \frac{1}{2}, -\frac{1}{2}, 0\right) &= \sqrt{3} \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 1 \\ \frac{1}{2} & -\frac{1}{2} & 0 \end{pmatrix}, \\
 C\left(\frac{1}{2}, \frac{1}{2}, 1 \mid -\frac{1}{2}, \frac{1}{2}, 0\right) &= \sqrt{3} \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 1 \\ -\frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}.
 \end{aligned}$$

Both these $3j$ -symbols correspond to the same entry in Table 16.1, and the symmetry rules give each the value $+1/\sqrt{6}$. Therefore, both these Clebsch-Gordan coefficients evaluate to $\sqrt{3}/\sqrt{6}$, or, as expected, $1/\sqrt{2}$.

For $|J = 0, M = 0\rangle$, we have, again calling on Eq. (16.47),

$$\begin{aligned}
 C\left(\frac{1}{2}, \frac{1}{2}, 0 \mid \frac{1}{2}, -\frac{1}{2}, 0\right) &= \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \end{pmatrix}, \\
 C\left(\frac{1}{2}, \frac{1}{2}, 0 \mid -\frac{1}{2}, \frac{1}{2}, 0\right) &= \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}.
 \end{aligned}$$

Again these $3j$ -symbols both correspond to the same tabulated entry (with value $1/\sqrt{2}$), but this time the symmetry rules cause them to have the respective values $+1/\sqrt{2}$ and $-1/\sqrt{2}$, in agreement with our explicit evaluation. ■

Example 16.2.3 COUPLING OF p AND d ELECTRONS

As most physics students know, a p state is an angular momentum eigenstate with $l = 1$ (so m can be 1, 0, or -1). The three normalized functions constituting its multiplet are often denoted p_+ , p_0 , and p_- . A d state has $l = 2$; we denote the five normalized members of its

multiplet d_{+2} , d_+ , d_0 , d_- , and d_{-2} . The m_1, m_2 basis has 15 members; grouped according to their M values, they consist of

$$\begin{array}{llll}
 M = +3 & p_+d_{+2} & & \\
 M = +2 & p_+d_+ & p_0d_{+2} & \\
 M = +1 & p_+d_0 & p_0d_+ & p_-d_{+2} \\
 M = 0 & p_+d_- & p_0d_0 & p_-d_+ \\
 M = -1 & p_+d_{-2} & p_0d_- & p_-d_0 \\
 M = -2 & p_0d_{-2} & p_-d_- & \\
 M = -3 & p_-d_{-2} & &
 \end{array}$$

This is the same coupling of angular momenta $j = 1$ and $j = 2$ that was introduced at the beginning of the subsection entitled Vector Model, but we are now illustrating how to carry out the coupling computations using Clebsch-Gordan coefficients and $3j$ -symbols. From this diagram, we expect one multiplet with $J = 3$, which in atomic spectroscopy is denoted F (multiparticle orbital angular momentum states are designated using upper-case letters); one with $J = 2$ (called D), and one with $J = 1$ (called P). Our plan is to construct these using the $3j$ -symbols given in Table 16.1.

We start by writing, in the notation $|J, M\rangle$, the members of the F multiplet with $M \geq 1$ in terms of Clebsch-Gordan coefficients (those for $M < 1$ do not raise important new points):

$$\begin{aligned}
 |3, 3\rangle &= C(1, 2, 3|1, 2, 3)p_+d_{+2}, \\
 |3, 2\rangle &= C(1, 2, 3|1, 1, 2)p_+d_+ + C(1, 2, 3|0, 2, 2)p_0d_{+2}, \\
 |3, 1\rangle &= C(1, 2, 3|1, 0, 1)p_+d_0 + C(1, 2, 3|0, 1, 1)p_0d_+ + C(1, 2, 3|-1, 2, 1)p_-d_{+2}.
 \end{aligned}$$

The D and P multiplet members for $M \geq 1$ are

$$\begin{aligned}
 |2, 2\rangle &= C(1, 2, 2|1, 1, 2)p_+d_+ + C(1, 2, 2|0, 2, 2)p_0d_{+2}, \\
 |2, 1\rangle &= C(1, 2, 2|1, 0, 1)p_+d_0 + C(1, 2, 2|0, 1, 1)p_0d_+ + C(1, 2, 2|-1, 2, 1)p_-d_{+2}, \\
 |1, 1\rangle &= C(1, 2, 1|1, 0, 1)p_+d_0 + C(1, 2, 1|0, 1, 1)p_0d_+ + C(1, 2, 1|-1, 2, 1)p_-d_{+2}.
 \end{aligned}$$

We then express the Clebsch-Gordan coefficients in terms of $3j$ -symbols. Doing just a representative few, using Eq. (16.47) and then the symmetry rules, Eqs. (16.48) and (16.49),

$$\begin{aligned}
 C(1, 2, 3|1, 2, 3) &= +\sqrt{7} \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & -3 \end{pmatrix} = 1, \\
 C(1, 2, 2|1, 1, 2) &= -\sqrt{5} \begin{pmatrix} 1 & 2 & 2 \\ 1 & 1 & -2 \end{pmatrix} = +\sqrt{5} \begin{pmatrix} 1 & 2 & 2 \\ 1 & -2 & 1 \end{pmatrix} = \sqrt{\frac{1}{3}}, \\
 C(1, 2, 1|-1, 2, 1) &= \sqrt{3} \begin{pmatrix} 1 & 2 & 1 \\ -1 & 2 & -1 \end{pmatrix} = \sqrt{3} \begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & -2 \end{pmatrix} = \sqrt{\frac{3}{5}}.
 \end{aligned}$$

Substituting these and other Clebsch-Gordan coefficients into the formulas for $|J, M\rangle$, we obtain the final results:

$$\begin{aligned}
 |3, 3\rangle &= p_+ d_{+2}, \\
 |3, 2\rangle &= \sqrt{\frac{1}{3}} p_0 d_{+2} + \sqrt{\frac{2}{3}} p_+ d_+, \\
 |3, 1\rangle &= \sqrt{\frac{1}{15}} p_- d_{+2} + \sqrt{\frac{8}{15}} p_0 d_+ + \sqrt{\frac{2}{5}} p_+ d_0, \\
 |2, 2\rangle &= -\sqrt{\frac{2}{3}} p_0 d_{+2} + \sqrt{\frac{1}{3}} p_+ d_+, \\
 |2, 1\rangle &= -\sqrt{\frac{1}{3}} p_- d_{+2} - \sqrt{\frac{1}{6}} p_0 d_+ + \sqrt{\frac{1}{2}} p_+ d_0, \\
 |1, 1\rangle &= \sqrt{\frac{3}{5}} p_- d_{+2} - \sqrt{\frac{3}{10}} p_0 d_+ + \sqrt{\frac{1}{10}} p_+ d_0.
 \end{aligned}$$

The reader may verify that states of the same M but different J have the required orthogonality. It is also easy to check that all these $|J, M\rangle$ states are normalized. ■

Exercises

- 16.2.1** Derive recursion relations for Clebsch-Gordan coefficients. Use them to calculate $C(11J | m_1 m_2 M)$ for $J = 0, 1, 2$.

Hint. Use the known matrix elements of $J_+ = J_{1+} + J_{2+}$, J_{i+} , and $\mathbf{J}^2 = (\mathbf{J}_1 + \mathbf{J}_2)^2$, etc.

- 16.2.2** Defining $(Y_l \chi)_J^M$ by the formula

$$(Y_l \chi)_J^M = \sum C(l \frac{1}{2} J | m_l m_s M) Y_{lm_l} \chi_{m_s},$$

where $\chi_{\pm 1/2}$ are the spin up and down eigenfunctions of $\sigma_3 = \sigma_z$, show that $(Y_l \chi)_J^M$ is a J, M eigenfunction.

- 16.2.3** Find the (j, m) states of a p electron ($l = 1$), in which the orbital angular momentum of the electron is coupled to its spin angular momentum ($s = 1/2$) to form states whose conventional labelings are $^2p_{1/2}$ and $^2p_{3/2}$. The notation is of the general form $^{2s+1}(\text{symbol})_j$, where “symbol” is that indicating the l value (i.e., $s, p, \dots$).

- 16.2.4** Repeat [Exercise 16.2.3](#) for $l = 1, s = 3/2$. Apply the conventional labels to the j, m states.

- 16.2.5** A deuterium atom consists of a proton, a neutron, and an electron. Each of these particles has spin $1/2$. The coupling of these three spins can produce J values of $3/2$ and $1/2$. We consider here only states with no orbital angular momentum.

- (a) Show that these J, M states consist of one quartet ($J = 3/2$) and **two** linearly independent doublets ($J = 1/2$).

Hint. Make a vector-model diagram.

- (b) One way to analyze this problem is to couple the spins of the proton and neutron to form a nuclear triplet or singlet, and then to couple the resultant nuclear spin to the electron spin. Find the states that are obtained in this way (designate the single-particle states $p_\alpha, p_\beta, n_\alpha, n_\beta, e_\alpha, e_\beta$).
- (c) Another way to analyze this problem is to couple the spins of the proton and electron to form an atomic triplet or singlet, and then to couple that resultant to the neutron spin. Find the states that result from this coupling scheme.
- (d) Show that the coupling schemes of parts (b) and (c) span the same Hilbert space.

Note. The actual interaction energies among these angular momenta cause the scheme of part (b) to be the better way of treating this problem (the triplet nuclear state is substantially the more stable), and the system actually looks like a spin-1 deuterium nucleus plus an electron.

16.3 SPHERICAL TENSORS

We have already seen that the set of spherical harmonics of given l transforms within itself under rotations. We now pursue this idea more formally. In Chapter 3 we saw that rotations could be characterized by the 3×3 unitary transformation matrices that transform a set of coordinates (their basis) into the new set corresponding to the rotation. These matrices could be viewed as second-rank tensors, but because they are restricted to rotational transformations, they are also known as **spherical tensors**.

We now wish to consider spherical tensors that transform more general sets of objects under rotation, and in particular those spherical tensors that have spherical harmonics as bases. Our new spherical tensors will then have dimensions other than 3×3 ; in fact, they must exist at all the sizes that correspond to sets of angular momentum eigenfunctions. Because we have already observed that a set of angular momentum eigenfunctions of a given J cannot be decomposed into subsets that transform only among themselves under rotation, we go one step further and call our spherical tensors **irreducible**.

Continuing for general angular momentum eigenfunctions $|L, M\rangle$, which we assume are representable in 3-D space as spherical harmonics or objects built from them by angular momentum coupling, we write the following defining equation for the spherical tensor describing the effect of a coordinate rotation $\mathbf{R}$ on $|L, M\rangle$:

$$\mathbf{R}|L, M\rangle = \sum_{M'} D_{M'M}^L(\mathbf{R})|L, M'\rangle. \quad (16.51)$$

If the $|L, M\rangle$ are actually spherical harmonics (and not more complicated objects that resulted from angular momentum coupling), Eq. (16.51) can also be written as

$$Y_l^m(\mathbf{R}\Omega) = \sum_{m'} D_{m'm}^l(\mathbf{R})Y_l^{m'}(\Omega). \quad (16.52)$$

Because we do not need to become embroiled in the details of the action of $\mathbf{R}$ on the coordinates, we have simply replaced (θ, φ) by the generic symbol Ω and have written $\mathbf{R}\Omega$ to indicate the coordinates (θ', φ') that describe the point that was labeled (θ, φ) in the unrotated system. For any given l , $D_{m'm}^l(\mathbf{R})$ can be regarded as an element of a square

matrix of dimension $2l + 1$ with rows and columns labeled by indices m' and m whose ranges are $(-l, \dots, +l)$, not the more customary sequence starting from 1. The $D_{m'm}^l(\mathbf{R})$ are unitary, since they describe a transformation between two orthonormal sets. Because of their early exploitation by Eugene Wigner, they are sometimes called Wigner matrices. There is an extensive literature (see Additional Readings) on relationships satisfied by the $D_{m'm}^l(\mathbf{R})$ and on formulas for their evaluation. A related topic included in this book is the formula, Eq. (3.37), giving the transformation of the basis x, y, z by a rotation through Euler angles α, β, γ .

Addition Theorem

Equation (16.52) can be used to establish important rotational invariance properties. For example, consider a quantity A defined as

$$A = \sum_m Y_l^m(\Omega_1)^* Y_l^m(\Omega_2), \quad (16.53)$$

where Ω_1 and Ω_2 are two unrelated sets of angular coordinates. We apply a rotation $\mathbf{R}$ to the coordinate system, denoting the result $\mathbf{R}A$, and evaluating the right-hand side using Eq. (16.52):

$$\mathbf{R}A = \sum_m \left(\sum_{\mu} D_{\mu m}^l(\mathbf{R}) Y_l^{\mu}(\Omega_1) \right)^* \left(\sum_{\nu} D_{\nu m}^l(\mathbf{R}) Y_l^{\nu}(\Omega_2) \right). \quad (16.54)$$

We now reorder the summations in Eq. (16.54), and, in the second line of Eq. (16.55), use the fact that $\mathbf{D}$ is unitary to change $\mathbf{D}^*$ to the transpose of $\mathbf{D}^{-1}$, thereby leading to the simplification in the third line. We have

$$\begin{aligned} \mathbf{R}A &= \sum_{\mu\nu} \left(\sum_m D_{\mu m}^l(\mathbf{R})^* D_{\nu m}^l(\mathbf{R}) \right) Y_l^{\mu}(\Omega_1)^* Y_l^{\nu}(\Omega_2) \\ &= \sum_{\mu\nu} \left(\sum_m \left[\mathbf{D}^l(\mathbf{R})^{-1} \right]_{m\mu} \left[\mathbf{D}^l(\mathbf{R}) \right]_{\nu m} \right) Y_l^{\mu}(\Omega_1)^* Y_l^{\nu}(\Omega_2) \\ &= \sum_{\mu\nu} \delta_{\mu\nu} Y_l^{\mu}(\Omega_1)^* Y_l^{\nu}(\Omega_2) = \sum_{\mu} Y_l^{\mu}(\Omega_1)^* Y_l^{\mu}(\Omega_2) = A. \end{aligned} \quad (16.55)$$

This shows that A is rotationally invariant, and is the starting point for an explanation of why a totally occupied atomic subshell (particles occupying all m values for a given l) leads to a spherically symmetric overall distribution.

The rotational invariance of A makes it easier for us to actually evaluate it, because we can choose to do so at a coordinate orientation for which the computation is relatively simple. Let's rotate the coordinates to place Ω_1 in the polar direction (so now $\theta_1 = 0$), and the θ value of Ω_2 in the rotated coordinates will be equal to the angle χ between the Ω_1 and Ω_2 directions, which is not affected by a coordinate rotation. In this new set of coordinates, $Y_l^m(\Omega_1)$ is $Y_l^m(0, \varphi)$ and is given, according to Eq. (15.148), as

$$Y_l^m(\Omega_1) = \sqrt{\frac{2l+1}{4\pi}} \delta_{m0}.$$

The summation in Eq. (16.53) therefore reduces to its $m = 0$ term, and the only Ω_2 contribution we need is $Y_l^0(\chi, \varphi_2)$. But because $m = 0$, this Y does not actually depend on φ_2 , and has the unambiguous value, from Eq. (15.137),

$$Y_l^0(\chi, \varphi_2) = \sqrt{\frac{2l+1}{4\pi}} P_l(\cos \chi).$$

These results enable us to obtain

$$A = \frac{2l+1}{4\pi} P_l(\cos \chi), \quad (16.56)$$

which, because of the rotational invariance, remains true whether or not the coordinate system was rotated. Inserting the original formula for A , and solving Eq. (16.56) for $P_l(\cos \chi)$, we obtain the **spherical harmonic addition theorem**,

$$P_l(\cos \chi) = \frac{4\pi}{2l+1} \sum_m Y_l^m(\Omega_1)^* Y_l^m(\Omega_2), \quad (16.57)$$

where χ is the angle between the directions Ω_1 and Ω_2 .

Example 16.3.1 ANGLE BETWEEN TWO VECTORS

A useful special case of the addition theorem is for $l = 1$, for which $P_1(\cos \chi) = \cos \chi$. Then, writing $\Omega_i \equiv \theta_i, \varphi_i$, and evaluating all the spherical harmonics on the right-hand side of Eq. (16.57), we have

$$\begin{aligned} \cos \chi &= \frac{1}{2} \left(\sin \theta_1 e^{-i\varphi_1} \right)^* \left(\sin \theta_2 e^{-i\varphi_2} \right) + \cos \theta_1 \cos \theta_2 \\ &\quad + \frac{1}{2} \left(-\sin \theta_1 e^{i\varphi_1} \right)^* \left(-\sin \theta_2 e^{i\varphi_2} \right) \\ &= \cos \theta_1 \cos \theta_2 + \frac{1}{2} \sin \theta_1 \sin \theta_2 \left(e^{i(\varphi_1 - \varphi_2)} + e^{i(\varphi_2 - \varphi_1)} \right). \end{aligned} \quad (16.58)$$

This reduces to the standard formula for the angle χ between directions (θ_1, φ_1) and (θ_2, φ_2) :

$$\cos \chi = \cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2 \cos(\varphi_2 - \varphi_1). \quad (16.59)$$

■

Spherical Wave Expansion

An important application of the addition theorem is the **spherical wave expansion**, which states

$$e^{i\mathbf{k} \cdot \mathbf{r}} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l i^l j_l(kr) Y_l^m(\Omega_k)^* Y_l^m(\Omega_r) \quad (16.60)$$

$$= 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l i^l j_l(kr) Y_l^m(\Omega_k) Y_l^m(\Omega_r)^*. \quad (16.61)$$

Here k and r are the magnitudes of $\mathbf{k}$ and $\mathbf{r}$, and Ω_k, Ω_r denote their respective angular coordinates. The two forms shown are equivalent because a change in the sign of m changes each harmonic to its complex conjugate (possibly with both harmonics undergoing a sign change). The quantity $j_l(kr)$ is a spherical Bessel function. This formula is particularly useful because it expresses the plane wave on its left-hand side as a series of spherical waves. This conversion is useful in scattering problems in which a plane wave, incident upon a scattering center, produces outgoing spherical waves with different spherical-harmonic (called **partial-wave**) components.

To establish Eq. (16.61), we write $\mathbf{k} \cdot \mathbf{r}$ as $kr \cos \chi$, where χ is the angle between $\mathbf{k}$ and $\mathbf{r}$, and then expand $\exp(ikr \cos \chi)$ as a series of Legendre polynomials:

$$e^{ikr \cos \chi} = \sum_{l=0}^{\infty} c_l P_l(\cos \chi), \quad (16.62)$$

with the coefficients c_l given by

$$c_l = \frac{2l+1}{2} \int_{-1}^1 e^{ikrt} P_l(t) dt. \quad (16.63)$$

We now recognize the integral in Eq. (16.63) as proportional to an integral representation of j_l that was the topic of Exercise 15.2.26 and which we repeat here:

$$j_l(x) = \frac{i^{-l}}{2} \int_{-1}^1 e^{ixt} P_l(t) dt. \quad (16.64)$$

This permits us to evaluate c_l , obtaining

$$c_l = (2l+1)i^l j_l(kr).$$

Inserting this expression for c_l into Eq. (16.62) and replacing $P_l(\cos \chi)$ in that equation by its equivalent as given by the addition theorem, Eq. (16.57), we have the desired verification of Eq. (16.61).

Laplace Spherical Harmonic Expansion

Another application of the addition theorem is to the Laplace expansion, where in Chapter 15 we found that the inverse distance between points $\mathbf{r}_1$ and $\mathbf{r}_2$ could be expanded in Legendre polynomials:

$$\frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} = \sum_{l=0}^{\infty} \frac{r_{<}^l}{r_{>}^{l+1}} P_l(\cos \chi). \quad (16.65)$$

Here $\mathbf{r}_1$ and $\mathbf{r}_2$ are measured from a common origin, with respective magnitudes r_1 and r_2 ; χ is the angle between $\mathbf{r}_1$ and $\mathbf{r}_2$. We define $r_{>}$ and $r_{<}$ as, respectively, the larger and

the smaller of r_1 and r_2 . If we now insert the addition theorem, we bring this expansion to the form

$$\frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} = \sum_{l=0}^{\infty} \frac{4\pi}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} \sum_{m=-l}^l Y_l^m(\Omega_1)^* Y_l^m(\Omega_2), \quad (16.66)$$

where Ω_1 and Ω_2 are the angular coordinates of $\mathbf{r}_1$ and $\mathbf{r}_2$ in a coordinate system of arbitrary orientation.

Example 16.3.2 SPHERICAL GREEN'S FUNCTION

An explicit expansion of the Green's function for the 3-D Laplace equation may be obtained by considering its defining equation

$$\nabla_1^2 G(\mathbf{r}_1, \mathbf{r}_2) = \delta(r_1 - r_2) \frac{\delta(\Omega_1 - \Omega_2)}{r_1^2}, \quad (16.67)$$

where we have written ∇_1 to remind the reader that it acts only on $\mathbf{r}_1$. Also, note that on the right-hand side the factor $1/r_1^2$ is inserted to adjust the angular delta function to unit scale; it could equally well have been written $1/r_2^2$ because of the presence also of $\delta(r_1 - r_2)$.

We now insert into Eq. (16.67), the following general expansion for $G(\mathbf{r}_1, \mathbf{r}_2)$:

$$G(\mathbf{r}_1, \mathbf{r}_2) = \sum_{lm} \sum_{l'm'} g_{ll'mm'}(r_1, r_2) Y_{l'}^{m'}(\Omega_1) Y_l^m(\Omega_2)^*,$$

and the expansion of Exercise 16.3.9 for the angular delta function:

$$\delta(\Omega_1 - \Omega_2) = \sum_{lm} Y_l^m(\Omega_1) Y_l^m(\Omega_2)^*.$$

We also write the Laplacian in the form

$$\nabla_1^2 = \frac{\partial^2}{\partial r_1^2} + \frac{2}{r_1} \frac{\partial}{\partial r_1} - \frac{\mathbf{L}_1^2}{r_1^2},$$

where $\mathbf{L}_1$ operates only on functions of Ω_1 .

We next take scalar products of the resulting expanded equation with all possible spherical harmonics of both Ω_1 and Ω_2 , in addition taking note that $Y_l^m(\Omega_1)$ is an eigenfunction of $\mathbf{L}_1^2$ with eigenvalue $l(l+1)$. We find that many terms cancel, so the scalar products lead, for each l and m , to the following result:

$$\left[\frac{d^2}{dr_1^2} + \frac{2}{r_1} \frac{d}{dr_1} - l(l+1) \right] g_l(r_1, r_2) = \delta(r_1 - r_2). \quad (16.68)$$

We have collapsed the original four indices of $g_{ll'mm'}(r_1, r_2)$ into the single index l because all instances of Eq. (16.68) with $l \neq l'$ or $m \neq m'$ vanish, and g has the same value for all m .

Equation (16.68) is for each l an ODE which, with boundary conditions $g = 0$ at $r = 0$ and $r = \infty$, defines the **spherical Green's functions** we identified in Section 10.2. Since

the homogeneous equation corresponding to Eq. (16.68) has solutions r^l and r^{-l-1} , its Green's function must have the form

$$g(r_1, r_2) = A_l \frac{r_{<}^l}{r_{>}^{l+1}}, \quad (16.69)$$

with $A_l = -1/(2l+1)$, a result that can be obtained by application of Eq. (10.19).

Comparing Eq. (16.66) with the result for $G(\mathbf{r}_1, \mathbf{r}_2)$ obtained by using Eq. (16.69), we now have yet another way of verifying the result that is familiar from Coulomb's law:

$$G(\mathbf{r}_1, \mathbf{r}_2) = - \sum_{l=0}^{\infty} \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} \sum_{m=-l}^l Y_l^m(\Omega_1)^* Y_l^m(\Omega_2) \quad (16.70)$$

$$= - \frac{1}{4\pi} \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|}. \quad (16.71)$$

■

General Multipoles

We are now ready to return to the multipole expansion. Given a set of charges q_i at respective points $\mathbf{r}_i$, all located within a sphere of radius a centered at the origin of a spherical polar coordinate system, we now consider the calculation of the electrostatic potential $\psi(\mathbf{r})$ at points outside the sphere, i.e., at points $\mathbf{r}$ such that $r > a$. Our starting point is the Laplace expansion of $1/|\mathbf{r}_1 - \mathbf{r}_2|$ in the form presented as Eq. (16.66). Since for all r_i we have $r_i < r$, we can write

$$\begin{aligned} \psi(\mathbf{r}) &= \frac{1}{4\pi\epsilon_0} \sum_i q_i \sum_{l=0}^{\infty} \frac{4\pi}{2l+1} \frac{r_i^l}{r^{l+1}} \sum_{m=-l}^l Y_l^m(\theta_i, \varphi_i)^* Y_l^m(\theta, \varphi) \\ &= \frac{1}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{4\pi}{2l+1} \left[\sum_i q_i r_i^l Y_l^m(\theta_i, \varphi_i)^* \right] \frac{Y_l^m(\theta, \varphi)}{r^{l+1}}. \end{aligned} \quad (16.72)$$

We see that this substitution has caused the entire effect of the charges q_i to be localized into the expressions

$$M_l^m = \frac{4\pi}{2l+1} \sum_i q_i r_i^l Y_l^m(\theta_i, \varphi_i)^*, \quad (16.73)$$

so that the potential due to the q_i , for points farther from $r = 0$ than all the charges, assumes the compact form,

$$\psi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^l M_l^m \frac{Y_l^m(\theta, \varphi)}{r^{l+1}}. \quad (16.74)$$

Equation (16.74) is called the **multipole expansion**, and the M_l^m are known as the **multipole moments** of the charge distribution. At this point we note that different authors define the multipole moments with different scalings, making up the difference by the inclusion of an appropriate factor in their formulas corresponding to Eq. (16.74). One reason for the

variety of notations is that M_l^m as defined in Eq. (16.73), which leads to the simplest formulas, does not yield the low-order moments at their “traditional” scalings. For example, the monopole moment, M_0^0 , evaluates to $(4\pi)^{1/2}$ times the total charge, while M_1^0 , the z -component of the dipole moment, comes out as $(4\pi/3)^{1/2} \sum_i q_i z_i$.

Of more fundamental interest is the relation between the multipole moments and the Cartesian forms that can represent them. We proceed by considering the M_l^m that result from a unit charge placed at (x, y, z) . Using the Cartesian representations of the spherical harmonics given in Table 15.4, the first few M_l^m have the forms given here:

$$\begin{aligned}
 M_2^2 &= \left(\frac{3\pi}{10}\right)^{1/2} (x^2 - y^2 + 2ixy) \\
 M_1^1 &= -\left(\frac{2\pi}{3}\right)^{1/2} (x + iy) & M_2^1 &= -\left(\frac{3\pi}{40}\right)^{1/2} z(x + iy) \\
 M_0^0 &= (4\pi)^{1/2} & M_1^0 &= \left(\frac{4\pi}{3}\right)^{1/2} z & M_2^0 &= \left(\frac{4\pi}{5}\right)^{1/2} \left[\frac{2z^2 - x^2 - y^2}{2} \right] \\
 M_1^{-1} &= \left(\frac{2\pi}{3}\right)^{1/2} (x - iy) & M_2^{-1} &= \left(\frac{3\pi}{40}\right)^{1/2} z(x - iy) \\
 M_2^{-2} &= \left(\frac{3\pi}{10}\right)^{1/2} (x^2 - y^2 - 2ixy)
 \end{aligned}$$

The first point to note is that for any l value, the Cartesian representation of each M_l^m involves a homogeneous polynomial of combined degree l in x , y , and z . It is obviously necessary that the M_l^m of different m be linearly independent, and we see that for $l = 0$ and $l = 1$, the number of independent monomials is equal to $2l + 1$, the number of m values. Specifically, for $l = 0$ we have only the monomial 1, while for $l = 1$ we have x , y , and z . But for $l = 2$, there are six independent monomials (x^2 , y^2 , z^2 , xy , xz , yz), but only five values of m . The discrepancy is resolved by observing that one linear combination of these monomials, namely $r^2 = x^2 + y^2 + z^2$, remains invariant under all rotations of the coordinates, and it therefore has different symmetry properties than the five-dimensional space orthogonal to r^2 . In fact, r^2 has the same symmetry as M_0^0 , but has the wrong r dependence to contribute to a solution to the Laplace equation (and therefore to the potential of a charge distribution).

If we were to continue to $l = 3$, we would find that there are 10 linearly independent monomials of degree 3, but they divide into a group of seven functions (the space spanned by M_3^m) with an orthogonal complement (functions orthogonal to the first seven) of dimension 3. These three remaining functions have a rotational symmetry similar to M_1^m , but again with the wrong r dependence to contribute to the potential. This type of pattern continues to higher l , making logical the observation that a multipole moment of degree l (a “ 2^l -moment”) has only $2l + 1$ components, despite the fact that in general the space of homogeneous polynomials of degree l has a larger dimension.

The multipole expansion is useful for continuous distributions of charge in addition to the discrete charge sets we have considered up to this point. The generalization of Eq. (16.73) is

$$M_l^m = \frac{4\pi}{2l + 1} \int \rho(\mathbf{r}') (r')^{l+2} Y_l^m(\theta', \varphi')^* \sin \theta' dr' d\theta' d\varphi', \quad (16.75)$$

where $\rho(\mathbf{r})$ is the charge density. This expression will yield valid results when $\psi(\mathbf{r})$ is computed via Eq. (16.74) for r values greater than the largest r' for which $\rho(\mathbf{r})$ is nonzero.

Integrals of Three Spherical Harmonics

Our final spherical tensor application is to the integrals of three spherical harmonics (all of the same argument). These integrals arise in the evaluation of matrix elements of angle-dependent operators which themselves can be written in terms of spherical harmonics. While it is possible to evaluate some such integrals using the techniques illustrated in Eq. (15.152), a more general result is available. Note that this is not an angular momentum coupling problem of the type we considered in Section 16.2, because that section treated angular momenta with independent arguments that depended on different variables. Here we have a different and more specialized situation in which all three angular momentum functions have **the same argument**.

The formula we seek is most easily derived if we have access to values of some of the rotation coefficients $D_{m'm}^l$ (a.k.a. Wigner matrices) defined in Eq. (16.52). The coefficients we need can be easily deduced with the aid of the spherical harmonic addition theorem, so we start by establishing the following lemma (a **lemma** is a mathematical result needed to prove something else):

Lemma: Evaluation of $D_{m0}^l(\mathbf{R})$:

Writing first the spherical harmonic addition theorem, Eq. (16.57),

$$P_l(\cos \chi) = \frac{4\pi}{2l+1} \sum_m Y_l^m(\Omega_1)^* Y_l^m(\Omega_2),$$

where χ is the angle between the directions Ω_1 and Ω_2 , we replace its left-hand side by the equivalent form

$$P_l(\cos \chi) = \sqrt{\frac{4\pi}{2l+1}} Y_l^0(\chi, 0),$$

thereby reaching

$$Y_l^0(\chi, 0) = \sqrt{\frac{4\pi}{2l+1}} \sum_m Y_l^m(\Omega_1)^* Y_l^m(\Omega_2). \quad (16.76)$$

We now compare this expression with Eq. (16.52), which we write here in a notation designed to make the comparison more obvious:

$$Y_l^0(\mathbf{R}\Omega_2) = \sum_m D_{m0}^l(\mathbf{R}) Y_l^m(\Omega_2). \quad (16.77)$$

If we select $\mathbf{R}$ to be a rotation that converts Ω_1 to the polar direction, then $\mathbf{R}\Omega_2$ will be $(\chi, 0)$; note that $Y_l^0(\mathbf{R}\Omega_2)$ is independent of φ so we can set its φ coordinate to zero. Thus, the comparison of Eqs. (16.76) and (16.77) yields

$$D_{m0}^l(\mathbf{R}) = \sqrt{\frac{4\pi}{2l+1}} Y_l^m(\Omega_1)^*. \quad (16.78)$$

We remind the reader that $\mathbf{R}$ is a rotation that converts Ω_1 to the polar direction.

Equation (16.78) has been derived under the assumption that the quantities being rotationally transformed are spherical harmonics (and not more complicated angular-momentum functions such as might be obtained via angular-momentum coupling). However, it is possible to show that the result generalizes, without change, to any angular momentum functions of integer l . ■

We now continue toward the goal of this subsection, namely the evaluation of integrals involving three spherical harmonics. The result we seek involves products of spherical harmonics with the same argument, but our method of obtaining that result proceeds by considering the rotational behavior of an angular momentum coupling formula (i.e., a product involving spherical harmonics of different arguments). So we now look at a special case of Eq. (16.45),

$$Y_{l_1}^0(\Omega_1)Y_{l_2}^0(\Omega_2) = \sum_L C(l_1, l_2, L|0, 0, 0)|L, 0\rangle, \quad (16.79)$$

where $|j_1, m_1; j_2, m_2\rangle$ of Eq. (16.45) is the product of spherical harmonics with $m_1 = m_2 = 0$ shown on the left-hand side of Eq. (16.79); the $|J, M\rangle$ state of Eq. (16.45) is now $|L, 0\rangle$. We next apply a rotation $\mathbf{R}$ to Eq. (16.79), using Eqs. (16.51) and (16.52) to get

$$\sum_{m_1 m_2} D_{m_1 0}^{l_1}(\mathbf{R}) D_{m_2 0}^{l_2}(\mathbf{R}) Y_{l_1}^{m_1}(\Omega_1) Y_{l_2}^{m_2}(\Omega_2) = \sum_{L, \sigma} C(l_1, l_2, L|0, 0, 0) D_{\sigma 0}^L(\mathbf{R}) |L, \sigma\rangle. \quad (16.80)$$

Finally, we convert $|L, \sigma\rangle$ back to the m_1, m_2 basis, using Eq. (16.43):

$$\begin{aligned} \sum_{m_1 m_2} D_{m_1 0}^{l_1}(\mathbf{R}) D_{m_2 0}^{l_2}(\mathbf{R}) Y_{l_1}^{m_1}(\Omega_1) Y_{l_2}^{m_2}(\Omega_2) &= \sum_{L, \sigma} C(l_1, l_2, L|0, 0, 0) D_{\sigma 0}^L(\mathbf{R}) \\ &\times \sum_{m_1 m_2} C(l_1, l_2, L|m_1, m_2, \sigma) Y_{l_1}^{m_1}(\Omega_1) Y_{l_2}^{m_2}(\Omega_2). \end{aligned} \quad (16.81)$$

This relatively complicated equation must be satisfied for all values of Ω_1 and Ω_2 , which will only be possible if its two sides are equal for each set of m_1, m_2 values. We therefore have the set of simpler equations,

$$D_{m_1 0}^{l_1}(\mathbf{R}) D_{m_2 0}^{l_2}(\mathbf{R}) = \sum_{L, \sigma} C(l_1, l_2, L|0, 0, 0) C(l_1, l_2, L|m_1, m_2, \sigma) D_{\sigma 0}^L(\mathbf{R}), \quad (16.82)$$

satisfied separately for all values of the free parameters.

We are now ready to replace all the D^l in Eq. (16.82) by the result obtained in our lemma, Eq. (16.78). Since the rotation $\mathbf{R}$ is arbitrary, both in the lemma and in the present work, our use of Eq. (16.78) will produce some angular coordinates Ω that have nothing to do with the Ω_i we were previously using; the point that is important here is that because the same $\mathbf{R}$ occurs throughout Eq. (16.82), the application of Eq. (16.78) will everywhere

produce the same Ω . Substitution of the lemma result yields

$$\frac{4\pi}{\sqrt{(2l_1+1)(2l_2+1)}} Y_{l_1}^{m_1}(\Omega)^* Y_{l_2}^{m_2}(\Omega)^* = \sum_{L\sigma} C(l_1, l_2, L|0, 0, 0) C(l_1, l_2, L|m_1, m_2, \sigma) \sqrt{\frac{4\pi}{2L+1}} Y_L^\sigma(\Omega)^*.$$

Since the Y_l^m are the only potentially complex quantities appearing here, we may remove the complex conjugate signs by complex conjugating the entire equation. After other minor rearrangements and recognition of the fact that the only contributing σ value is $\sigma = m_1 + m_2$, we reach the final form

$$Y_{l_1}^{m_1}(\Omega) Y_{l_2}^{m_2}(\Omega) = \sum_L \sqrt{\frac{(2l_1+1)(2l_2+1)}{4\pi(2L+1)}} \times C(l_1, l_2, L|0, 0, 0) C(l_1, l_2, L|m_1, m_2, m_1+m_2) Y_L^{m_1+m_2}(\Omega). \quad (16.83)$$

At last we can meet the objective of this subsection. Multiplying both sides of Eq. (16.83) by some $Y_{l_3}^{m_3}(\Omega)^*$ and integrating in Ω over the angular space, we get

$$\begin{aligned} \langle Y_{l_3}^{m_3} | Y_{l_1}^{m_1} | Y_{l_2}^{m_2} \rangle &= \int_0^{2\pi} d\varphi \int_0^\pi \sin\theta d\theta Y_{l_3}^{m_3}(\theta, \varphi)^* Y_{l_1}^{m_1}(\theta, \varphi) Y_{l_2}^{m_2}(\theta, \varphi) \\ &= \sqrt{\frac{(2l_1+1)(2l_2+1)}{4\pi(2L+1)}} C(l_1, l_2, l_3|0, 0, 0) C(l_1, l_2, l_3|m_1, m_2, m_3). \end{aligned} \quad (16.84)$$

We do not have to include a Kronecker delta because the condition $m_3 = m_1 + m_2$ is taken care of by the fact that the Clebsch-Gordan coefficients vanish in the absence of this or any other condition needed for a nonzero result.

Some further insight can be obtained by considering the special case $m_1 = m_2 = m_3 = 0$ and writing the spherical harmonics in terms of Legendre polynomials. This brings us (after the substitution $t = \cos\theta$) to

$$\int_{-1}^1 P_{l_3}(t) P_{l_1}(t) P_{l_2}(t) dt = \frac{2}{2l_3+1} C(l_1, l_2, l_3|0, 0, 0)^2. \quad (16.85)$$

Since we know that the Legendre polynomial $P_l(t)$ of even l is an even function of t , while that of odd l is odd in t , we see from Eq. (16.85) that unless $l_1 + l_2 + l_3$ is even, the integral will vanish, telling us that $C(l_1, l_2, l_3|0, 0, 0)$ will only be nonzero if $l_1 + l_2 + l_3$ is even. In addition, if the product of any two of the $P_l(t)$ does not contain a power of t as large as the index of the third P_l , the integral will vanish due to the orthogonality of the Legendre functions. This observation translates into a **triangle condition**, namely that the integral will vanish unless $|l_1 - l_2| \leq l_3 \leq l_1 + l_2$. Since these are conditions on the Clebsch-Gordan coefficient $C(l_1, l_2, l_3|0, 0, 0)$, they apply also to the general integral formula, Eq. (16.84).

Summarizing, integrals of products of three spherical harmonics, evaluated in Eq. (16.84), will only be nonzero if the three following conditions are satisfied:

1. The l values satisfy the triangle condition $|l_1 - l_2| \leq l_3 \leq l_1 + l_2$.
2. The m values satisfy the condition $m_3 = m_1 + m_2$.
3. The sum of the l values, $l_1 + l_2 + l_3$, is even.

Exercises

- 16.3.1 For $l = 1$, Eq. (16.52) becomes

$$Y_1^m(\theta', \varphi') = \sum_{m'=-1}^1 D_{m'm}^1(\alpha, \beta, \gamma) Y_1^{m'}(\theta, \varphi).$$

Rewrite these spherical harmonics in Cartesian form. Show that the resulting Cartesian coordinate equations are equivalent to the Euler rotation matrix $\mathbf{A}(\alpha, \beta, \gamma)$, Eq. (3.37).

- 16.3.2 In proving the addition theorem, we assumed that $Y_l^k(\theta_1, \varphi_1)$ could be expanded in a series of $Y_l^m(\theta_2, \varphi_2)$, in which m varied from $-l$ to $+l$ but l was held fixed. What arguments can you develop to justify summing only over the upper index, m , and **not** over the lower index, l ?

Hints. One possibility is to examine the homogeneity of the Y_l^m , that is, Y_l^m may be expressed entirely in terms of the form $\cos^{l-p} \theta \sin^p \theta$, or $x^{l-p-s} y^p z^s / r^l$. Another possibility is to examine the behavior of the Legendre equation under rotation of the coordinate system.

- 16.3.3 An atomic electron with angular momentum l and magnetic quantum number m has a wave function

$$\psi(r, \theta, \varphi) = f(r) Y_l^m(\theta, \varphi).$$

Show that the sum of the electron densities in a given complete shell is spherically symmetric; that is, $\sum_{m=-l}^l \psi^*(r, \theta, \varphi) \psi(r, \theta, \varphi)$ is independent of θ and φ .

- 16.3.4 The potential of an electron at point $\mathbf{r}_e$ in the field of Z protons at points $\mathbf{r}_p$ is

$$\Phi = -\frac{e^2}{4\pi\epsilon_0} \sum_{p=1}^Z \frac{1}{|\mathbf{r}_e - \mathbf{r}_p|}.$$

Show that for r_e larger than all r_p , this may be written as

$$\Phi = -\frac{e^2}{4\pi\epsilon_0 r_e} \sum_{p=1}^Z \sum_{L,M} \left(\frac{r_p}{r_e}\right)^L \frac{4\pi}{2L+1} Y_L^M(\theta_p, \varphi_p)^* Y_L^M(\theta_e, \varphi_e).$$

How should Φ be written for $r_e < r_p$?

- 16.3.5 Two protons are **uniformly** distributed within the same spherical volume. If the coordinates of one element of charge are $(r_1, \theta_1, \varphi_1)$ and the coordinates of the other are $(r_2, \theta_2, \varphi_2)$ and r_{12} is the distance between them, the element of repulsion energy will be given by

$$d\psi = \rho^2 \frac{d\tau_1 d\tau_2}{r_{12}} = \rho^2 \frac{r_1^2 dr_1 \sin\theta_1 d\theta_1 d\varphi_1 r_2^2 dr_2 \sin\theta_2 d\theta_2 d\varphi_2}{r_{12}},$$

where

$$\rho = \frac{\text{charge}}{\text{volume}} = \frac{3e}{4\pi R^3} \quad \text{and} \quad r_{12}^2 = r_1^2 + r_2^2 - 2r_1r_2 \cos \gamma.$$

Here ρ is the charge density and γ is the angle between $\mathbf{r}_1$ and $\mathbf{r}_2$. Calculate the total electrostatic energy (of repulsion) of the two protons. This calculation is used in accounting for the mass difference in “mirror” nuclei, such as O^{15} and N^{15} .

$$\text{ANS.} \quad \frac{6}{5} \frac{e^2}{R}.$$

- 16.3.6** Each of the two $1s$ electrons in helium may be described by a hydrogenic wave function

$$\psi(\mathbf{r}) = \left(\frac{Z^3}{\pi a_0^3} \right)^{1/2} e^{-Zr/a_0}$$

in the absence of the other electron. Here Z , the atomic number, is 2. The symbol a_0 is the Bohr radius, $\hbar^2/me^2$. Find the mutual potential energy of the two electrons, given by

$$\int \psi^*(\mathbf{r}_1) \psi^*(\mathbf{r}_2) \frac{e^2}{|\mathbf{r}_1 - \mathbf{r}_2|} \psi(\mathbf{r}_1) \psi(\mathbf{r}_2) d^3r_1 d^3r_2.$$

$$\text{ANS.} \quad \frac{5e^2Z}{8a_0}.$$

- 16.3.7** The probability of finding a $1s$ hydrogen electron in a volume element $r^2 dr \sin \theta d\theta d\varphi$ is

$$\frac{1}{\pi a_0^3} e^{-2r/a_0} r^2 dr \sin \theta d\theta d\varphi,$$

where r is the distance of the electron from the nucleus. Find the electrostatic potential of this charge distribution at points $\mathbf{r}_1$, where you may **not** assume that $\mathbf{r}_1$ is on the polar axis of your coordinate system. Calculate the potential from

$$V(\mathbf{r}_1) = -\frac{e}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}_2)}{r_{12}} d^3r_2,$$

where $r_{12} = |\mathbf{r}_1 - \mathbf{r}_2|$. Expand r_{12} . Apply the Legendre polynomial addition theorem and show that the angular dependence of $V(\mathbf{r}_1)$ drops out.

$$\text{ANS.} \quad V(\mathbf{r}_1) = -\frac{e}{4\pi\epsilon_0} \left[\frac{1}{2r_1} \gamma \left(3, \frac{2r_1}{a_0} \right) + \frac{1}{a_0} \Gamma \left(2, \frac{2r_1}{a_0} \right) \right], \text{ where}$$

γ and Γ are incomplete gamma functions, Eq. (13.73).

- 16.3.8** A hydrogen electron in a $2p$ orbital has a charge distribution

$$\rho = -\frac{e}{64\pi a_0^5} r^2 e^{-r/a_0} \sin^2 \theta,$$

where $a_0 = \hbar^2/me^2$ is the Bohr radius, and r is the distance between the electron and the nucleus. Find the electrostatic potential energy for this atomic state.

16.3.9 (a) As a Laplace series and as an example of Eq. (5.27), show that

$$\delta(\Omega_1 - \Omega_2) = \sum_{l=0}^{\infty} \sum_{m=-l}^l Y_l^m(\theta_2, \varphi_2)^* Y_l^m(\theta_1, \varphi_1).$$

(b) Show also that this **same** representation of the Dirac delta function may be written as

$$\delta(\Omega_1 - \Omega_2) = \sum_{l=0}^{\infty} \frac{2l+1}{4\pi} P_l(\cos \gamma),$$

and identify γ . Now, if you can justify equating the summations over l **term by term**, you have an alternate derivation of the spherical harmonic addition theorem.

16.3.10 Verify

$$\begin{aligned} \text{(a)} \quad & \int Y_L^M(\theta, \varphi) Y_0^0(\theta, \varphi) Y_L^{M*}(\theta, \varphi) d\Omega = \frac{1}{\sqrt{4\pi}}, \\ \text{(b)} \quad & \int Y_L^M Y_1^0 Y_{L+1}^{M*} d\Omega = \sqrt{\frac{3}{4\pi}} \sqrt{\frac{(L+M+1)(L-M+1)}{(2L+1)(2L+3)}}, \\ \text{(c)} \quad & \int Y_L^M Y_1^1 Y_{L+1}^{M+1*} d\Omega = \sqrt{\frac{3}{8\pi}} \sqrt{\frac{(L+M+1)(L+M+2)}{(2L+1)(2L+3)}}, \\ \text{(d)} \quad & \int Y_L^M Y_1^1 Y_{L-1}^{M+1*} d\Omega = -\sqrt{\frac{3}{8\pi}} \sqrt{\frac{(L-M)(L-M-1)}{(2L-1)(2L+1)}}. \end{aligned}$$

These integrals were used in an investigation of the angular correlation of internal conversion electrons.

16.3.11 Show that

$$\begin{aligned} \text{(a)} \quad & \int_{-1}^1 x P_L(x) P_N(x) dx = \begin{cases} \frac{2(L+1)}{(2L+1)(2L+3)}, & N = L+1, \\ \frac{2L}{(2L-1)(2L+1)}, & N = L-1, \end{cases} \\ \text{(b)} \quad & \int_{-1}^1 x^2 P_L(x) P_N(x) dx = \begin{cases} \frac{2(L+1)(L+2)}{(2L+1)(2L+3)(2L+5)}, & N = L+2, \\ \frac{2(2L^2+2L-1)}{(2L-1)(2L+1)(2L+3)}, & N = L, \\ \frac{2L(L-1)}{(2L-3)(2L-1)(2L+1)}, & N = L-2. \end{cases} \end{aligned}$$

16.3.12 Since $xP_n(x)$ is a polynomial (of degree $n+1$), it may be represented by the Legendre series

$$xP_n(x) = \sum_{s=0}^{\infty} a_s P_s(x).$$

- (a) Show that $a_s = 0$ for $s < n-1$ and $s > n+1$.
- (b) Calculate a_{n-1} , a_n , and a_{n+1} and show that you have reproduced the recurrence relation, Eq. (15.18).

Note. This argument may be put in a general form to demonstrate the existence of a three-term recurrence relation for any of our complete sets of orthogonal polynomials:

$$x\varphi_n = a_{n+1}\varphi_{n+1} + a_n\varphi_n + a_{n-1}\varphi_{n-1}.$$

16.4 VECTOR SPHERICAL HARMONICS

Maxwell's equations lead naturally to applications involving a vector Helmholtz equation for the vector potential $\mathbf{A}$, and various classical and quantum-mechanical problems in this area are usefully attacked by introducing **vector spherical harmonics**. Our first step in this direction will be to recognize that a set of unit vectors can be thought of as a spherical tensor of rank 1 and can be discussed in terms of the angular momentum formalism. We will later (in Chapter 17) pursue rotational symmetry in greater depth; for our present purposes it suffices to confirm the relationship between rotations in 3-D space and angular momentum operators.

A Spherical Tensor

We consider here vectors in 3-D space, of the form $\mathbf{u} = u_x \hat{\mathbf{e}}_x + u_y \hat{\mathbf{e}}_y + u_z \hat{\mathbf{e}}_z$, but, unlike our practice in Chapter 3, we will permit the u_j to be complex, and use the complex scalar product $\langle \mathbf{u} | \mathbf{u} \rangle^{1/2}$ as a measure of the magnitude of $\mathbf{u}$. If we restrict the vectors $\mathbf{u}$ to be of unit length, they satisfy the conditions necessary to be identified as spherical tensors of rank 1.

We now introduce operators K_i defined by the following matrices:

$$K_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad K_2 = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}, \quad K_3 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (16.86)$$

The reader can easily verify that these matrices satisfy the angular momentum commutation rules, and in fact describe the result of applying the angular momentum operator $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, where $\mathbf{p} = -i\nabla$, to the basis x, y, z . We next calculate

$$\mathbf{K}^2 = K_1^2 + K_2^2 + K_3^2 = 2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

showing that all members of the basis are eigenvectors of $\mathbf{K}^2$, with eigenvalue 2, which is $k(k+1)$ with $k=1$. All members of our basis therefore have one unit of some abstract sort of angular momentum (often referred to as **spin**), and we can obtain a set of eigenvectors with values of an index m that can have values $+1$, 0 , and -1 . By diagonalizing the matrix K_3 , we find its eigenvectors to be

$$\mathbf{k}_1 = \begin{pmatrix} -1/\sqrt{2} \\ -i/\sqrt{2} \\ 0 \end{pmatrix}, \quad \mathbf{k}_0 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{k}_{-1} = \begin{pmatrix} 1/\sqrt{2} \\ -i/\sqrt{2} \\ 0 \end{pmatrix}. \quad (16.87)$$

While in principle the signs of these eigenvectors are arbitrary, they have been chosen here to agree with the Condon-Shortley phase convention.

Vector Coupling

The vector spherical harmonics are now defined as the quantities that result from the coupling of ordinary spherical harmonics and the vectors $\mathbf{e}_m$ to form states of definite J (the resultant of the orbital angular momentum of the spherical harmonic and the one unit possessed by the $\mathbf{e}_m$). It is customary to label the vector spherical harmonics to show both the L value from the ordinary (scalar) harmonic and the M value (the eigenvalue of J_z). Thus, the vector spherical harmonic will have three indices: J , L , and M . From the general formula for angular-momentum coupling, Eq. (16.43), we have

$$\mathbf{Y}_{JLM}(\theta, \varphi) = \sum_{mm'} C(L, 1, J | mm' M) Y_L^m(\theta, \varphi) \hat{\mathbf{e}}_{m'}. \quad (16.88)$$

Remember that M is M_J , not the m value of Y_L^m , and that $\hat{\mathbf{e}}_{m'}$ are the angular momentum eigenfunctions given in Eq. (16.87).

Because Eq. (16.88) couples an angular momentum L with one of magnitude $k=1$, the L values in a vector spherical harmonic of given J are restricted to $J+1$, J , and $J-1$, a condition enforced by the values of the Clebsch-Gordan coefficients. Moreover, because the Clebsch-Gordan coefficients describe a unitary transformation, the obvious orthogonality of the states in the m, m' basis ($Y_L^m \hat{\mathbf{e}}_{m'}$) will cause the vector spherical harmonics also to be orthonormal:

$$\int \mathbf{Y}_{JLM}(\theta, \varphi) \cdot \mathbf{Y}_{J'L'M'}(\theta, \varphi) d\Omega = \delta_{JJ'} \delta_{LL'} \delta_{MM'}. \quad (16.89)$$

In addition, we can invert Eq. (16.89) using Eq. (16.45), reaching

$$Y_L^m(\theta, \varphi) \hat{\mathbf{e}}_{m'} = \sum_{JM} C(L, 1, J | mm' M) \mathbf{Y}_{JLM}. \quad (16.90)$$

The manipulation of expressions involving the vector spherical harmonics depends crucially on a few identities, of which perhaps the most important is the formula

$$\hat{\mathbf{r}} Y_L^M(\theta, \varphi) = - \left[\frac{L+1}{2L+1} \right]^{1/2} \mathbf{Y}_{L,L+1,M} + \left[\frac{L}{2L+1} \right]^{1/2} \mathbf{Y}_{L,L-1,M}. \quad (16.91)$$

To establish this formula, and at the same time to make its meaning more obvious, we start by noting that $\hat{\mathbf{r}}$ has a form that depends on the angular coordinates; specifically, it is

$$\hat{\mathbf{r}} = \frac{\mathbf{r}}{\sin \theta \cos \varphi \hat{\mathbf{e}}_x + \sin \theta \sin \varphi \hat{\mathbf{e}}_y + \cos \theta \hat{\mathbf{e}}_z}.$$

For our present purposes, it is more convenient to rearrange this to the form

$$\hat{\mathbf{r}} = \sin \theta \left(\frac{e^{i\varphi} \hat{\mathbf{e}}_{-1} - e^{-i\varphi} \hat{\mathbf{e}}_{+1}}{\sqrt{2}} \right) + \cos \theta \hat{\mathbf{e}}_0. \quad (16.92)$$

It is now clear that in order to prove Eq. (16.91) we must show that each $\hat{\mathbf{e}}_m$ has the same coefficient on both sides of the equation. Taking first the coefficient of $\hat{\mathbf{e}}_0$, the left-hand side of Eq. (16.91) yields, after use of Eq. (16.92),

$$\begin{aligned} \cos \theta Y_L^M(\theta, \varphi) &= \left[\frac{(l-m+1)(l+m+1)}{(2l+1)(2l+3)} \right]^{1/2} Y_{l+1}^m \\ &\quad + \left[\frac{(l-m)(l+m)}{(2l-1)(2l+1)} \right]^{1/2} Y_{l-1}^m, \end{aligned} \quad (16.93)$$

a result previously exhibited as Eq. (15.150). The $\hat{\mathbf{e}}_0$ terms from the right-hand side of Eq. (16.91) consist of

$$\begin{aligned} & - \left[\frac{L+1}{2L+1} \right]^{1/2} C(L+1, 1, L|M, 0, M) Y_{L+1}^M \hat{\mathbf{e}}_0 \\ & + \left[\frac{L}{2L+1} \right]^{1/2} C(L-1, 1, L|M, 0, M) Y_{L-1}^M \hat{\mathbf{e}}_0. \end{aligned}$$

The Clebsch-Gordan coefficients appearing here have the values

$$\begin{aligned} C(L+1, 1, L|M, 0, M) &= - \left[\frac{(L+M+1)(L-M+1)}{(L+1)(2L+3)} \right]^{1/2}, \\ C(L-1, 1, L|M, 0, M) &= \left[\frac{L^2 - M^2}{L(2L-1)} \right]^{1/2}. \end{aligned}$$

These data permit confirmation of the $\mathbf{e}_0$ terms of Eq. (16.91). The terms in $\mathbf{e}_{+1}$ and $\mathbf{e}_{-1}$ can also be shown consistent; the formulas needed for that purpose are Eqs. (15.151) and (15.152).

Another useful formula, which can be obtained by using Eq. (16.91) to simplify the radial component when the gradient operator is applied to the form $f(r)Y_L^M(\theta, \varphi)$, is

$$\begin{aligned} \nabla \left[f(r) Y_L^M(\theta, \varphi) \right] &= - \left[\frac{L+1}{2L+1} \right]^{1/2} \left[\frac{\partial}{\partial r} - \frac{L}{r} \right] f(r) \mathbf{Y}_{L, L+1, M}(\theta, \varphi) \\ &\quad + \left[\frac{L}{2L+1} \right]^{1/2} \left[\frac{\partial}{\partial r} + \frac{L+1}{r} \right] f(r) \mathbf{Y}_{L, L-1, M}(\theta, \varphi). \end{aligned} \quad (16.94)$$

Under coordinate inversion the vector spherical harmonics transform as

$$\begin{aligned}\mathbf{Y}_{L,L+1,M}(\theta', \varphi') &= (-1)^{L+1} \mathbf{Y}_{L,L+1,M}(\theta, \varphi), \\ \mathbf{Y}_{L,L-1,M}(\theta', \varphi') &= (-1)^{L+1} \mathbf{Y}_{L,L-1,M}(\theta, \varphi), \\ \mathbf{Y}_{LLM}(\theta', \varphi') &= (-1)^L \mathbf{Y}_{LLM}(\theta, \varphi),\end{aligned}\tag{16.95}$$

where

$$\theta' = \pi - \theta \quad \varphi' = \pi + \varphi.$$

Starting from Eqs. (16.91) and (16.94), a number of formulas can be derived for the divergence and curl of vector spherical harmonics. These formulas include the following:

$$\nabla \cdot [f(r) \mathbf{Y}_{L,L+1,M}(\theta, \varphi)] = - \left[\frac{L+1}{2L+1} \right]^{1/2} \left[\frac{df(r)}{dr} + \frac{L+2}{r} f(r) \right] Y_L^M(\theta, \varphi),\tag{16.96}$$

$$\nabla \cdot [f(r) \mathbf{Y}_{L,L-1,M}(\theta, \varphi)] = \left[\frac{L}{2L+1} \right]^{1/2} \left[\frac{df(r)}{dr} - \frac{L-1}{r} f(r) \right] Y_L^M(\theta, \varphi),\tag{16.97}$$

$$\nabla \cdot [f(r) \mathbf{Y}_{LLM}(\theta, \varphi)] = 0,\tag{16.98}$$

$$\nabla \times [f(r) \mathbf{Y}_{L,L+1,M}(\theta, \varphi)] = i \left[\frac{L}{2L+1} \right]^{1/2} \left[\frac{df(r)}{dr} + \frac{L+2}{r} f(r) \right] \mathbf{Y}_{LLM},\tag{16.99}$$

$$\begin{aligned}\nabla \times [f(r) \mathbf{Y}_{LLM}(\theta, \varphi)] &= i \left(\frac{L}{2L+1} \right)^{1/2} \left[\frac{df(r)}{dr} - \frac{L}{r} f(r) \right] \mathbf{Y}_{L,L+1,M}(\theta, \varphi), \\ &\quad + i \left[\frac{L+1}{2L+1} \right]^{1/2} \left[\frac{df(r)}{dr} + \frac{L+1}{r} f(r) \right] \mathbf{Y}_{L,L-1,M},\end{aligned}\tag{16.100}$$

$$\nabla \times [f(r) \mathbf{Y}_{L,L-1,M}(\theta, \varphi)] = i \left[\frac{L+1}{2L+1} \right]^{1/2} \left[\frac{df(r)}{dr} - \frac{L-1}{r} f(r) \right] \mathbf{Y}_{LLM}(\theta, \varphi).\tag{16.101}$$

For a complete derivation of Eqs. (16.96) to (16.101) we refer to the literature.³ These relations play an important role in the partial wave expansion of classical and quantum electrodynamics.

The definitions of the vector spherical harmonics given here are dictated by convenience, primarily in quantum mechanical calculations, in which the angular momentum is a significant parameter. Further examples of the usefulness and power of the vector spherical harmonics will be found in Blatt and Weisskopf, in Morse and Feshbach, and in Jackson (all in Additional Readings).

In closing, we note that

³E. H. Hill, Theory of vector spherical harmonics, *Am. J. Phys.* **22**: 211 (1954). Note that Hill assigns phases in accordance with the Condon-Shortley phase convention. In Hill's notation, $\mathbf{X}_{LM} = \mathbf{Y}_{LLM}$, $\mathbf{V}_{LM} = \mathbf{Y}_{L,L+1,M}$, $\mathbf{W}_{LM} = \mathbf{Y}_{L,L-1,M}$.

- Vector spherical harmonics are developed from coupling L units of orbital angular momentum and one unit of spin angular momentum.
- An extension, coupling L units of orbital angular momentum and two units of spin angular momentum to form **tensor** spherical harmonics, is presented by Mathews.⁴
- The major application of tensor spherical harmonics is in the investigation of gravitational radiation.

Exercises

16.4.1 Construct the $l = 0, m = 0$ and $l = 1, m = 0$ vector spherical harmonics.

$$\begin{aligned} \text{ANS. } \mathbf{Y}_{010} &= -\hat{\mathbf{r}}(4\pi)^{-1/2} \\ \mathbf{Y}_{000} &= 0 \\ \mathbf{Y}_{120} &= -\hat{\mathbf{r}}(2\pi)^{-1/2} \cos \theta - \hat{\boldsymbol{\theta}}(8\pi)^{-1/2} \sin \theta \\ \mathbf{Y}_{110} &= \hat{\boldsymbol{\phi}}(3/8\pi)^{1/2} \sin \theta \\ \mathbf{Y}_{100} &= \hat{\mathbf{r}}(4\pi)^{-1/2} \cos \theta - \hat{\boldsymbol{\theta}}(4\pi)^{-1/2} \sin \theta. \end{aligned}$$

16.4.2 Verify that the parity of $\mathbf{Y}_{LL+1M}$ is $(-1)^{L+1}$, that of $\mathbf{Y}_{LLM}$ is $(-1)^L$, and that of $\mathbf{Y}_{LL-1M}$ is $(-1)^{L+1}$. What happened to the M -dependence of the parity?

Hint. $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\phi}}$ have odd parity; $\hat{\boldsymbol{\theta}}$ has even parity (compare with Exercise 3.10.25).

16.4.3 Verify the orthonormality of the vector spherical harmonics $\mathbf{Y}_{JLMJ}$.

16.4.4 Jackson's *Classical Electrodynamics* (see Additional Readings) defines $\mathbf{Y}_{LLM}$ by the equation

$$\mathbf{Y}_{LLM}(\theta, \varphi) = \frac{1}{\sqrt{L(L+1)}} \mathbf{L} Y_L^M(\theta, \varphi),$$

in which the angular momentum operator $\mathbf{L}$ is given by

$$\mathbf{L} = -i(\mathbf{r} \times \nabla).$$

Show that this definition agrees with Eq. (16.88).

16.4.5 Show that

$$\sum_{M=-L}^L \mathbf{Y}_{LLM}^*(\theta, \varphi) \cdot \mathbf{Y}_{LLM}(\theta, \varphi) = \frac{2L+1}{4\pi}.$$

Hint. One way is to use Exercise 16.4.4 with $\mathbf{L}$ expanded in Cartesian coordinates and to apply raising and lowering operators.

⁴J. Mathews, Gravitational multipole radiation, *J. Soc. Ind. Appl. Math.* **10**: 768 (1963).

16.4.6 Show that

$$\int \mathbf{Y}_{LLM} \cdot (\hat{\mathbf{r}} \times \mathbf{Y}_{LLM}) d\Omega = 0.$$

The integrand represents an interference term in electromagnetic radiation that contributes to angular distributions but not to total intensity.

Additional Readings

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